Langevin Coupling of Oriented Disks

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1 Rotational Langevin equation for a 2D hexagonal disk array

We consider a two-dimensional triangular lattice of identical rigid disks of radius a. The disk centres are assumed to be fixed on the lattice, and each disk j can only rotate about its own centre. The corresponding angular coordinate and angular velocity are

$$\theta_j = \theta_j(t), \qquad \omega_j = \dot{\theta}_j. \tag{1.1}$$

The nearest-neighbour lattice spacing is $\ell = 2a + h_0$, where h_0 is the mean gap width between neighbouring disks. We work in the lubrication limit,$h_0 \ll a$, and assume creeping flow, so that the fluid in the thin gaps is described by Stokes flow and the lubrication approximation.

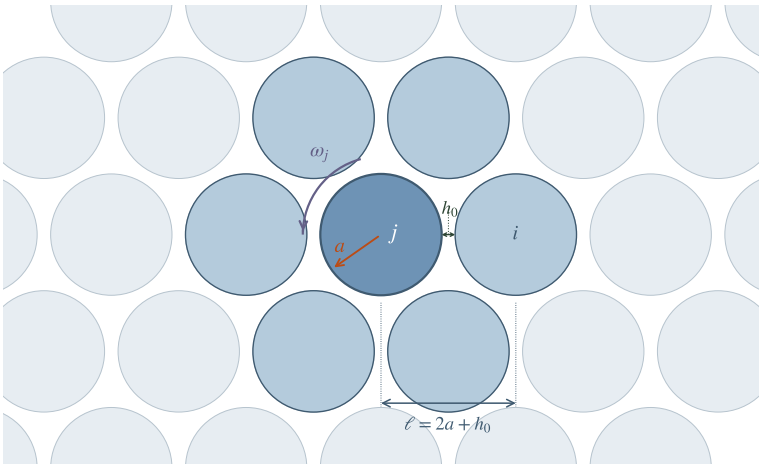


Figure 1: Schematic of the two-dimensional triangular lattice of rigid disks. The central disk j and its six nearest neighbours are highlighted. The lattice spacing is $\ell = 2a + h_0$, where a is the disk radius and h_0 is the mean gap width. Each disk rotates about its own centre with angular velocity ω_j.

1.1 Lubrication torque between two disks

Consider two neighbouring disks i and j. Let Γ_{ij} denote the thin-gap contact region between them, and let s be a coordinate along this gap. The local gap height is $h_{ij}(s)$. At a point on the gap, the surface velocity of disk j due to rotation is $\mathbf{u}_j(\mathbf{x}) = \omega_j\,\hat{\mathbf{z}} \times (\mathbf{x} - \mathbf{r}_j)$. For equal disks and nearest-neighbour geometry, the relative tangential surface velocity across the gap is proportional to the *sum* of the two angular velocities,

$$\Delta u_{ij}(s) \simeq a\big(\omega_i + \omega_j\big), \tag{1.2}$$

because the two facing surface points lie on the same side of their respective centres and their velocities add: for a co-rotating pair $\omega_i = \omega_j = \omega$ the facing surfaces slide past one another at $2a\omega$, whereas a counter-rotating pair would roll without slipping. In the lubrication limit, the shear stress transmitted through the gap is of the form

$$\tau_{ij}(s) \simeq \eta \frac{\Delta u_{ij}(s)}{h_{ij}(s)} = \eta a \frac{\omega_i + \omega_j}{h_{ij}(s)}, \tag{1.3}$$

where η is the dynamic viscosity. The resulting hydrodynamic torque on disk j from the gap with disk i is obtained by integrating the shear stress over the lubrication region,

$$T_{ij}^{\mathrm{lub}} = -a \int_{\Gamma_{ij}} \tau_{ij}(s)\,\mathrm{d}s = -\eta a^2 (\omega_i + \omega_j) \int_{\Gamma_{ij}} \frac{\mathrm{d}s}{h_{ij}(s)}. \tag{1.4}$$

It is convenient to define Γ_{ij} the dimensionless-in-time coupling coefficient

$$T_{ij}^{\mathrm{lub}} = -I\,\Gamma_{ij}\,(\omega_i + \omega_j), \qquad \Gamma_{ij} = \frac{\eta a^2}{I}\int_{\Gamma_{ij}} \frac{\mathrm{d}s}{h_{ij}(s)}, \tag{1.5}$$

where I is the moment of inertia of one disk per unit thickness. For a uniform hexagonal lattice, all six nearest-neighbour gaps are equivalent, so $\Gamma_{ij} = \Gamma$ for all nearest neighbours i, j .

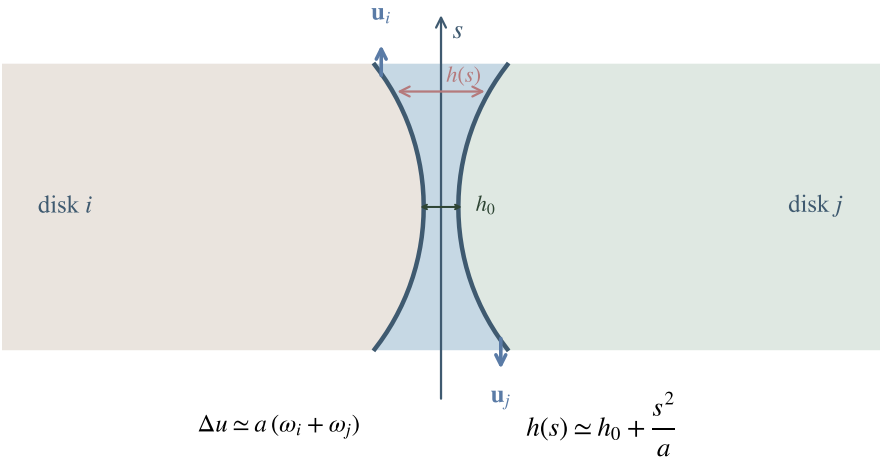


Figure 2: Enlarged view of the lubrication gap between two neighbouring disks i and j. The local gap height is $h(s) \simeq h_0 + s^2/a$. The relative tangential surface velocity is $\Delta u \simeq a(\omega_i + \omega_j)$, and the shear stress in the gap is $\tau \simeq \eta\Delta u/h(s)$. The integration region Γ_{ij} along the gap is indicated.

1.2 The effective Langevin equation

To begin with, the angular momentum balance for disk j is

$$I\dot{\omega}_j = -\gamma_{\mathrm{bulk}}\omega_j + \sum_{i\in\mathcal{N}(j)} T_{ij}^{\mathrm{lub}} + T_j^{\mathrm{st}}(t), \tag{1.6}$$

where γ_{bulk} is the (bulk) rotational damping from the surrounding fluid, $\mathcal{N}(j)$ denotes the six nearest neighbours of j, and $T_j^{\mathrm{st}}(t)$ is the stochastic torque from thermal fluid fluctuations. Substituting the lubrication torque for the six nearest neighbours gives

$$\dot{\omega}_j = -\frac{\gamma_{\mathrm{bulk}}}{I}\omega_j - \sum_{i\in\mathcal{N}(j)} \Gamma_{ij}(\omega_j + \omega_i) + \frac{1}{I}T_j^{\mathrm{st}}(t) = -\left(\frac{\gamma_{\mathrm{bulk}}}{I} + 6\Gamma\right)\omega_j - \Gamma\sum_{\langle i,j\rangle}\omega_i + \frac{1}{I}T_j^{\mathrm{st}}(t), \tag{1.7}$$

where $\sum_{\langle i,j\rangle}\omega_i$ denotes the sum over the six nearest neighbours of j. Rearranging,

$$\dot{\omega}_j + \left(\frac{\gamma_{\mathrm{bulk}}}{I} + 6\Gamma\right)\omega_j + \Gamma\sum_{\langle i,j\rangle}\omega_i = \frac{1}{I}T_j^{\mathrm{st}}(t). \tag{1.8}$$

This has exactly the proposed form in later sections

$$\boxed{\frac{\mathrm{d}\omega_j}{\mathrm{d}t} + \gamma_0\omega_j - c\Lambda_j(t) = -\gamma\sum_{\langle i,j\rangle}\omega_i,} \tag{1.9}$$

provided that the effective coefficients are identified as

$$\gamma_0 = \frac{\gamma_{\mathrm{bulk}}}{I} + 6\Gamma, \qquad \gamma = \Gamma = \frac{\eta a^2}{I} \int_{\Gamma_{ij}} \frac{\mathrm{d}s}{h_{ij}(s)}, \qquad c\Lambda_j(t) = \frac{1}{I} T_j^{\mathrm{st}}(t). \tag{1.10}$$

In particular, in such model the neighbouring disks are *anti*-coupled in the sense that the lubrication torque opposes their relative rotation, and the single-particle damping of a non-uniform rotation pattern is enhanced. Here $\Lambda_j(t)$ is white noise independent on each disk and satisfies

$$\langle \Lambda_i(t) \rangle = 0, \qquad \langle \Lambda_i(t) \Lambda_j(t') \rangle = \delta(t - t') \delta_{ij}. \tag{1.11}$$

1.3 Estimated coefficient forms

For equal disks with a small gap h_0, the gap height near the closest approach is approximately $h(s) \simeq h_0 + \frac{s^2}{a}$. Thus the lubrication integral and coefficient Γ have respective scaling

$$\int_\Gamma \frac{\mathrm{d}s}{h(s)} \sim \int_{-\infty}^{\infty} \frac{\mathrm{d}s}{h_0 + s^2/a} = \pi \sqrt{\frac{a}{h_0}}, \qquad \Gamma \sim \frac{\eta a^2}{I} \sqrt{\frac{a}{h_0}}. \tag{1.12}$$

For a disk with $I = \frac{1}{2} \rho \pi a^4$ per unit thickness, this becomes

$$\Gamma \sim \frac{2\eta}{\rho a^2} \sqrt{\frac{a}{h_0}} = \frac{2\nu}{a^2} \sqrt{\frac{a}{h_0}}, \tag{1.13}$$

where $\nu = \eta / \rho$ is the kinematic viscosity. The bulk damping coefficient γ_{bulk} depends on the outer fluid and boundary conditions. For an isolated rotating disk in a Stokes fluid, $\gamma_{\mathrm{bulk}} \sim \eta a^2$ up to a dimensionless prefactor. Thus

$$\gamma_0 = \frac{\gamma_{\mathrm{bulk}}}{I} + 6\Gamma \sim \frac{2\nu}{a^2} \left(C_{\mathrm{bulk}} + 6 \sqrt{\frac{a}{h_0}} \right), \tag{1.14}$$

where C_{bulk} is an $O(1)$ geometric constant. The lubrication calculation only fixes their scaling and physical origin. In practice, the exact numerical prefactors in γ_0, γ, and c should be obtained by fitting to experiments or numerical simulations.

2 Mode Analysis Beyond Mean Field: Effective Damping and Short-Range Correlations

We now consider the Langevin equations justified by the lubrication limit, written in the form (1.9) with the explicit sign implied by the identification (1.10),

$$\frac{d\theta_j}{dt} = \omega_j, \qquad \frac{d\omega_j}{dt} = -\gamma_0 \omega_j - \gamma \sum_{\langle i,j \rangle} \omega_i + c\, \Lambda_j(t), \tag{2.1}$$

where $\gamma > 0$ and $\Lambda_j(t)$ is independent Gaussian white noise satisfying (1.11). The initial conditions are $\theta_j(0) = \omega_j(0) = 0$.

2.1 Mean-field effective damping

A simple mean-field approximation replaces the neighbour sum by its local average, $\sum_{\langle i,j\rangle}\omega_i \longrightarrow z\,\bar{\omega}_j$, for the periodic triangular lattice $z=6$. Then the single-site equation becomes

$$\frac{d\omega_j}{dt} = -(\gamma_0+6\gamma)\omega_j + c\,\Lambda_j(t). \tag{2.2}$$

Thus the mean-field effective damping is

$$\boxed{\Gamma_{\mathrm{MFT}} = \Gamma_0 \equiv \gamma_0 + 6\gamma = \gamma_0 + 6\Gamma,} \tag{2.3}$$

which is the same Γ_0 as in (1.10). Equation (2.2) is an Ornstein–Uhlenbeck process. Its stationary variance is

$$\langle\omega_j^2\rangle_{\mathrm{st}} = \frac{c^2}{2\Gamma_0}, \qquad \mathrm{Var}(\theta_j(t)) \simeq \frac{c^2}{\Gamma_0^2}\,t, \qquad t \gg \Gamma_0^{-1}. \tag{2.4}$$

Hence θ_j is an unbounded zero-mean diffusive variable, and its variance grows linearly in time. This is the (unfaithful) mean-field result.

2.2 Fourier analysis on the triangular lattice

To go beyond the mean-field approximation, we Fourier transform the linear ω-equation. Let

$$\omega_{\mathbf{k}}(t) = \frac{1}{\sqrt{M}}\sum_j e^{-i\mathbf{k}\cdot\mathbf{r}_j}\omega_j(t), \qquad \theta_{\mathbf{k}}(t) = \frac{1}{\sqrt{M}}\sum_j e^{-i\mathbf{k}\cdot\mathbf{r}_j}\theta_j(t). \tag{2.5}$$

For a triangular lattice with lattice constant a, the six nearest-neighbour vectors are

$$\boldsymbol{\delta}_1 = a(1,0), \quad \boldsymbol{\delta}_2 = a\left(\frac{1}{2},\frac{\sqrt{3}}{2}\right), \quad \boldsymbol{\delta}_3 = a\left(-\frac{1}{2},\frac{\sqrt{3}}{2}\right), \tag{2.6}$$

and their negatives. The Fourier transform of the coupling term is $\sum_{\langle i,j\rangle}\omega_i \xrightarrow{\mathrm{FT}} S(\mathbf{k})\,\omega_{\mathbf{k}}$, where the triangular-lattice structure factor is

$$S(\mathbf{k}) = 2\left[\cos(k_x a) + 2\cos\left(\frac{k_x a}{2}\right)\cos\left(\frac{\sqrt{3}\,k_y a}{2}\right)\right]. \tag{2.7}$$

Therefore each Fourier mode evolves independently as

$$\boxed{\frac{d\omega_{\mathbf{k}}}{dt} = -\Gamma(\mathbf{k})\omega_{\mathbf{k}} + c\,\Lambda_{\mathbf{k}}(t), \qquad \Gamma(\mathbf{k}) = \gamma_0 + \gamma S(\mathbf{k}).} \tag{2.8}$$

At $\mathbf{k}=0$, $S(0)=6$, so $\Gamma(0)=\gamma_0+6\gamma=\gamma_0+6\Gamma=\Gamma_0$, which coincides with the MFT result. Since $\gamma>0$, the k^2 term is *negative* and Γ is *largest* at $\mathbf{k}=0$: the uniform (long-wavelength) mode is the *fastest* one, because all disks co-rotate and every gap is sheared. The *slowest* modes are instead the anti-aligned ones, whose six neighbours add up with a negative sign; these occur at the K points, where $S(K)=-3$. Meanwhile, the mean-field approximation uses only $\Gamma_0=\Gamma(0)$, that is, the fastest rate. For each mode, the stationary variance of $\omega_{\mathbf{k}}$ is

$$\langle|\omega_{\mathbf{k}}|^2\rangle_{\mathrm{st}} = \frac{c^2}{2\Gamma(\mathbf{k})}. \tag{2.9}$$

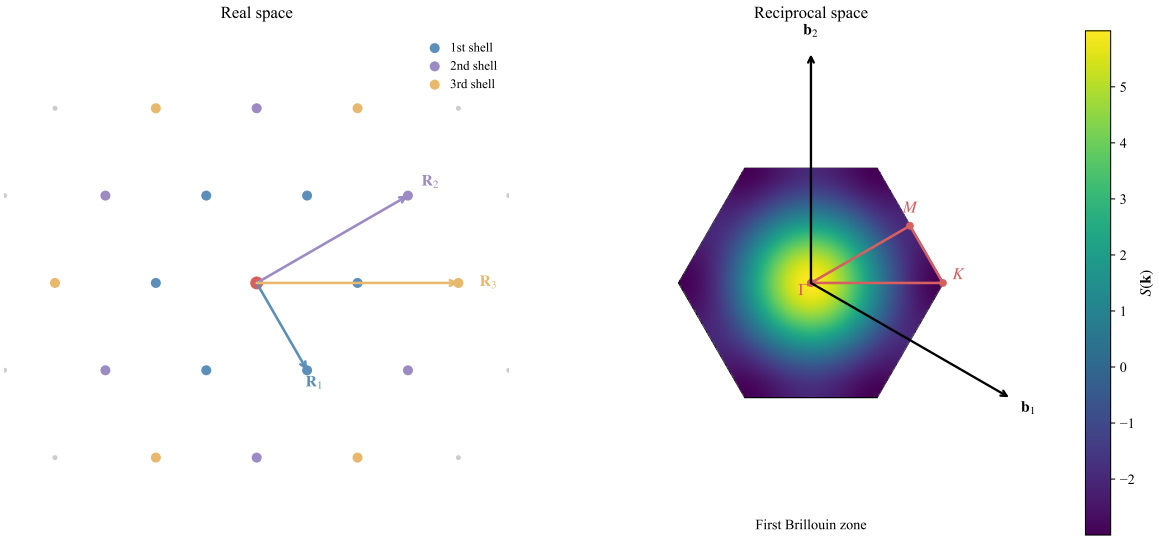


Figure 3: Left: real-space triangular lattice with the first five neighbour shells. The six nearest-neighbour vectors $\boldsymbol{R}_1 \sim \boldsymbol{R}_3$ are shown. Right: reciprocal space showing the hexagonal Brillouin zone, the reciprocal lattice vectors $\mathbf{b}_1, \mathbf{b}_2$, and the high-symmetry path Γ-M-K-Γ.

Since $\theta_{\mathbf{k}}(t) = \int_0^t \omega_{\mathbf{k}}(s)\, ds$, the equal-time correlation in Fourier space with $t \gg \Gamma(\mathbf{k})^{-1}$ is proportional to t

$$C_\theta(\mathbf{k}, t) \equiv \langle |\theta_{\mathbf{k}}(t)|^2 \rangle \simeq \frac{c^2\, t}{\Gamma(\mathbf{k})^2}. \tag{2.10}$$

The precise statement behind (2.10), and the regime in which it holds, are given in Sec. 3.1; the leading correction is discussed there and does not affect any conclusion of this section.

2.3 Inverse transform and short-range correlations

The inverse of the lattice transform (2.10) is an integral over the first Brillouin zone. On a large lattice the allowed wavevectors are dense enough that the zone may be treated as continuous; in this model the two descriptions agree to 5×10^{-5} for 2048^2 sites (Sec. 3.4), and the continuum form is the one that gives the correct behaviour at large separations. Writing $A_{\mathrm{BZ}} = (2\pi)^2 / A_{\mathrm{cell}}$ with $A_{\mathrm{cell}} = \sqrt{3}a^2/2$, the equal-time correlation is

$$\boxed{C_\theta(\mathbf{r}, t) = \frac{c^2 t}{A_{\mathrm{BZ}}} \int_{\mathrm{BZ}} \frac{d^2k}{(2\pi)^2}\, \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\Gamma(\mathbf{k})^2}\,,} \tag{2.11}$$

with $\Gamma(\mathbf{k}) = \gamma_0 + \gamma S(\mathbf{k})$. Equivalently, and this is the form used in the numerics,

$$C_\theta^{\mathrm{Pea}}(\mathbf{r}, t) = \frac{C_\theta(\mathbf{r}, t)}{C_\theta(\mathbf{0}, t)} = \frac{\int_{\mathrm{BZ}} d^2k\, e^{i\mathbf{k}\cdot\mathbf{r}}/\Gamma(\mathbf{k})^2}{\int_{\mathrm{BZ}} d^2k\, 1/\Gamma(\mathbf{k})^2}, \tag{2.12}$$

in which the common prefactor $c^2 t / A_{\mathrm{BZ}}(2\pi)^2$ cancels, so that the Pearson correlation depends on the dynamics only through the dimensionless ratio $\mu = \gamma/\gamma_0$. Equation (2.12) is the form used for the shell table and the correlation figure below; it is a single exact result in the weak coupling limit, and the result relevant to a strong coupling is developed in Sec. 3.3.

There is an essential remark here. The Brillouin-zone integral in (2.11) is finite as it stands, because $1/\Gamma(\mathbf{k})^2$ is bounded ($\Gamma \geq \Gamma_{\min} > 0$) and the zone is compact, so no regularisation is needed. The point

$\mathbf{k} = 0$ carries no weight in the integral, and therefore no spurious uniform contribution appears. This is why the correlation obtained from (2.11) decays to zero at large $\mathbf{r}$, as a physical correlation must; we verify this explicitly on shells 1 to 14 in Sec. 3.4. By contrast, a *discrete* sum over the allowed wavevectors of a finite sample assigns weight $1/N$ to the single $\mathbf{k} = 0$ mode and therefore contributes an $\mathbf{r}$-independent term of order $1/N$ which does not decay; that term is an artefact of the finite sample rather than a property of the model, and it vanishes in the continuum description of a large lattice.

2.4 Real-space correlations

We now compute the equal-time correlation

$$C_\theta(\mathbf{r}, t) = \langle \theta_{\mathbf{r}}(t)\theta_{\mathbf{0}}(t)\rangle \tag{2.13}$$

on the first five neighbour shells of the triangular lattice, from the Brillouin-zone integral (2.11) with $S(\mathbf{k})$ given in Eq. (2.7). The whole $\mathbf{k}$-dependence of the integrand is known analytically and the integrals are evaluated directly; therefore no small-k or long-wavelength approximation is needed anywhere in this paper.

The first five shells of the triangular lattice are

Shell	m^2	Distance squared	Number of sites
1	1	a^2	6
2	3	$3a^2$	6
3	4	$4a^2$	6
4	7	$7a^2$	12
5	9	$9a^2$	6

with $\mathbf{R} = a(m_1 + m_2/2, \sqrt{3}m_2/2)$, so that $r/a = 1, \sqrt{3}, 2, \sqrt{7}, 3$.

Table 1 lists C_θ^{Pea} on these shells for several values of μ, and Fig. 4 displays the same data with significant features. With positive μ, the sign of correlation alternates along the shells, due to the hexagonal anisotropy of $\Gamma(\mathbf{k})$. In addition, the tail alternates with rapidly decreasing magnitude and stronger coupling makes $\Gamma(\mathbf{k})$ more strongly $\mathbf{k}$-dependent.

μ	Γ_0	shell 1	shell 2	shell 3	shell 4	shell 5
-0.10	0.40	$+0.30133$	$+0.10914$	$+0.07813$	$+0.03124$	$+0.01938$
-0.05	0.70	$+0.11879$	$+0.01906$	$+0.01148$	$+0.00223$	$+0.00105$
-0.02	0.88	$+0.04263$	$+0.00260$	$+0.00140$	$+0.00013$	$+0.00004$
0.00	1.00	$+0.00002$	-0.00002	-0.00001	$+0.00002$	$+0.00000$
$+0.02$	1.12	-0.03775	$+0.00221$	$+0.00101$	-0.00007	-0.00002
$+0.05$	1.30	-0.08748	$+0.01265$	$+0.00505$	-0.00110	-0.00021
$+0.10$	1.60	-0.15730	$+0.04522$	$+0.01337$	-0.00721	-0.00022

Table 1: Pearson correlation C_θ^{Pea} on the first five shells from the Brillouin-zone integral (2.11), for a perfect triangular lattice with $a = 1$ and $\gamma_0 = 1$. The kernel is $1/\Gamma(\mathbf{k})^2$, i.e. the transient window (2.10); the exact stationary kernel $1/\Gamma$ that replaces it once the slowest mode has saturated is treated in Sec. 3.3, and the same shells are tabulated there for the stationary kernel. The values are Pearson-normalised, hence independent of c, t and the overall scale of Γ; they were obtained by extrapolating the Brillouin-zone integral in the grid spacing, on the same recipe as Table 3. The lubrication case is $\mu > 0$.

The numerical method used to evaluate (2.11), together with its verification and convergence tests, is described in Sec. 3.4. The correlation decays to zero at large $\mathbf{r}$, because the continuous Brillouin-zone integral contains no weight at the single point $\mathbf{k} = 0$, whereas a discrete sum over the allowed wavevectors

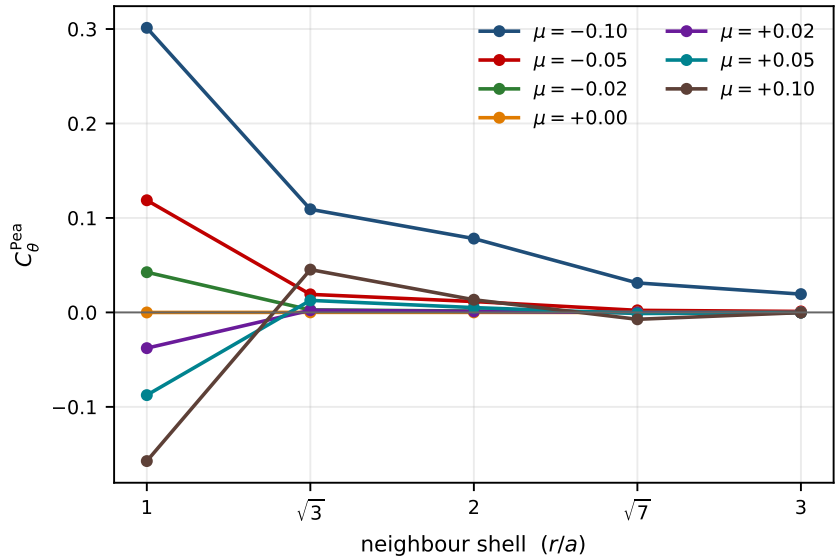


Figure 4: Pearson correlation C_θ^{Pea} on the first five neighbour shells ($r/a = 1, \sqrt{3}, 2, \sqrt{7}, 3$) from the Brillouin-zone integral (2.11), for several values of $\mu = \gamma/\gamma_0$. The sign of the nearest-neighbour correlation is the sign of μ, and the sequence alternates because of the hexagonal anisotropy of $\Gamma(\mathbf{k})$.

of a finite sample would assign that mode a weight $1/N$ and hence an $\mathbf{r}$-independent contribution. This is quantified in Sec. 3.4 and shown in Fig. 6.

It should be stressed that (2.11) is not yet the correlation at a finite observation time but the t-linear window of the exact mode-resolved variance (3.6). The complete treatment, together with the stationary kernel $1/\Gamma$ that replaces $1/\Gamma^2$ as soon as the slowest mode has saturated, is given in the next section.

3 The stationary correlation of the coupled array

Section 2.4 evaluated the equal-time correlation under weak coupling condition, with the kernel $1/\Gamma(\mathbf{k})^2$. Notwithstanding, the physics of interest for a colloidal system is the opposite one, in which the coupling γ is not small compared with the local damping γ_0 and the lubrication coupling then makes the mode relaxation rates strongly $\mathbf{k}$-dependent. For instance, the slowest rate $\Gamma_{\min} = \gamma_0 + \gamma S_{\min} = \gamma_0\,(1 - 3\mu)$ becomes much smaller, and the relaxation time $1/\Gamma_{\min}$ therefore grows without bound as the coupling approaches its limiting value, with $\Gamma_{\min} t \lesssim 1$ inaccessible to an experiment whose observation time is fixed. In the opposite regime, we will show in later calculation that $\Gamma_{\min} t \gg 1$, *every* mode has saturated and the kernel is $1/\Gamma(\mathbf{k})$, not $1/\Gamma(\mathbf{k})^2$. In this case, a larger coupling makes the stationary kernel simultaneously more relevant and more distinct from the transient one.

This section is organised around that stationary kernel. We first recall the exact mode-resolved variance, whose two windows are $1/\Gamma$ and $1/\Gamma^2$, and define the measured observable; we then state the stationary correlation and give its independent justification, its decay, and its behaviour as $\mu \to 1/3$; finally we describe the numerical evaluation and verify it.

3.1 Exact single-mode variance and the two windows

We still follows the analysis Fourier modes, which obey the Ornstein–Uhlenbeck equation in frequency space:

$$\frac{\mathrm{d}\omega_{\mathbf{k}}}{\mathrm{d}t} = -\Gamma(\mathbf{k})\,\omega_{\mathbf{k}} + c\,\Lambda_{\mathbf{k}}(t), \qquad \frac{\mathrm{d}\theta_{\mathbf{k}}}{\mathrm{d}t} = \omega_{\mathbf{k}}, \qquad \omega_{\mathbf{k}}(0) = \theta_{\mathbf{k}}(0) = 0, \tag{3.1}$$

with $\Gamma(\mathbf{k}) = \gamma_0 + \gamma S(\mathbf{k})$ from Eq. (2.8) and the noise normalisation $\langle \Lambda_{\mathbf{k}}(s)\Lambda_{\mathbf{k}}(s')\rangle = \delta(s-s')$. Integrating with the initial condition kept, the velocity correlation is

$$\langle \omega_{\mathbf{k}}(s)\,\omega_{\mathbf{k}}(s')\rangle = c^2 e^{-\Gamma(s+s')} \int_0^{\min(s,s')} e^{2\Gamma u}\,\mathrm{d}u = \frac{c^2}{2\Gamma}\left(e^{-\Gamma|s-s'|} - e^{-\Gamma(s+s')}\right). \tag{3.2}$$

The second term carries the initial condition $\omega_{\mathbf{k}}(0) = 0$ that makes (3.2) vanish at $s = s' = 0$ to give the accurate short time behaviour. Since $\theta_{\mathbf{k}}(t) = \int_0^t \omega_{\mathbf{k}}(s)\,\mathrm{d}s$, the equal-time variance of the angle is the double integral

$$\langle |\theta_{\mathbf{k}}(t)|^2\rangle = \frac{c^2}{2\Gamma}\int_0^t \mathrm{d}s \int_0^t \mathrm{d}s' \,\left(e^{-\Gamma|s-s'|} - e^{-\Gamma(s+s')}\right). \tag{3.3}$$

The first integral is elementary,

$$\int_0^t \mathrm{d}s \int_0^t \mathrm{d}s'\, e^{-\Gamma|s-s'|} = 2\int_0^t \mathrm{d}s \int_0^s \mathrm{d}s'\, e^{-\Gamma(s-s')} = \frac{2}{\Gamma}\int_0^t \mathrm{d}s\,\left(1 - e^{-\Gamma s}\right) = \frac{2}{\Gamma}\left(t - \frac{1-e^{-\Gamma t}}{\Gamma}\right), \tag{3.4}$$

and the second factorises,

$$\int_0^t \mathrm{d}s \int_0^t \mathrm{d}s'\, e^{-\Gamma(s+s')} = \left(\int_0^t e^{-\Gamma s}\,\mathrm{d}s\right)^2 = \frac{\left(1-e^{-\Gamma t}\right)^2}{\Gamma^2}. \tag{3.5}$$

Subtracting the two and expanding the square and substituting into (3.3),

$$\boxed{\langle |\theta_{\mathbf{k}}(t)|^2\rangle = \frac{c^2}{\Gamma(\mathbf{k})^2}\Big[\Gamma(\mathbf{k})\,t - 1 + e^{-\Gamma(\mathbf{k})t}\Big],} \tag{3.6}$$

which is exact for every mode and every time, with the limits

$$\langle |\theta_{\mathbf{k}}(t)|^2\rangle = \begin{cases} \dfrac{c^2 t^2}{2} + \mathcal{O}(t^3), & \Gamma t \ll 1 \quad \text{(damping not yet felt)},\\[2ex] \dfrac{c^2 t}{\Gamma}\left[1 - \dfrac{1}{\Gamma t} + \mathcal{O}\big(\tfrac{1}{\Gamma^2 t^2}\big)\right], & \Gamma t \gg 1 \quad \text{(saturated, diffusive)}. \end{cases} \tag{3.7}$$

Two remarks. First, the velocity field reaches its stationary variance on the single $\mathbf{k}$-independent time scale $\Gamma t \gg 1$, since $\langle |\omega_{\mathbf{k}}|^2\rangle = \frac{c^2}{2\Gamma}\left(1 - e^{-2\Gamma t}\right) \to c^2/(2\Gamma)$; by (3.7) the angle field has no such universal time, its behaviour being governed by $\Gamma(\mathbf{k})t$ mode by mode. This is why a measurement of ω equilibrates quickly while a shell-resolved measurement of θ keeps evolving, and it is why t must be kept finite. Second, the slowest modes carry the largest $\langle |\theta_{\mathbf{k}}|^2\rangle$, which scales as $1/\Gamma(\mathbf{k})$ for $\Gamma t \gg 1$ and as $1/\Gamma(\mathbf{k})^2$ for $\Gamma t \ll 1$. As S is a sum of six cosines, Γ is largest at $\mathbf{k} = \mathbf{0}$ for every μ (the uniform rotation shears every gap), and its minimum is at the K points, where $\Gamma(K) = \gamma_0(1 - 3\mu)$ softens as $\mu \to 1/3$.

3.2 Subtraction of the mean and the measured correlation

In practice, what is measured is not the bare angle field but the field with its spatial mean removed. Each frame defines

$$\delta\theta_j(t) = \theta_j(t) - \bar{\theta}(t), \qquad \bar{\theta}(t) = \frac{1}{N}\sum_j \theta_j(t), \tag{3.8}$$

and the shell-resolved correlator is built from $\delta\theta_i\,\delta\theta_j$. In Fourier space the subtraction is a projection, because $\theta_{\mathbf{0}}$ is the spatial mean up to the normalisation of the transform, so that

$$\delta\theta_{\mathbf{k}} = \theta_{\mathbf{k}} \quad (\mathbf{k} \neq \mathbf{0}), \qquad \delta\theta_{\mathbf{0}} = 0. \tag{3.9}$$

The zero mode is therefore removed by the definition of the observable, not by an additional modelling step. Because the modes are independent, the covariance of the fluctuation field at time t is

$$\langle \delta\theta_i\,\delta\theta_j\rangle(t) = \frac{c^2}{2N}\sum_{\mathbf{k}\neq\mathbf{0}} e^{i\mathbf{k}\cdot(\mathbf{r}_i-\mathbf{r}_j)}\,\frac{\Gamma(\mathbf{k})t - 1 + e^{-\Gamma(\mathbf{k})t}}{\Gamma(\mathbf{k})^2}, \tag{3.10}$$

where the factor $\frac{1}{2}$ is the usual real-field normalisation: for a zero-mean complex amplitude the real and imaginary parts are independent and each has variance $\frac{1}{2}\langle|\theta_{\mathbf{k}}|^2\rangle$.

Normalising by the value at $\mathbf{r} = \mathbf{0}$ and passing to the continuum description of the zone with the same convention as Eq. (2.11), $\sum_{\mathbf{k}\neq\mathbf{0}} \to (N/A_{\mathrm{BZ}})\int_{\mathrm{BZ}} \mathrm{d}^2k$ with $A_{\mathrm{BZ}} = (2\pi)^2/A_{\mathrm{cell}}$, so that the factors of N cancel, the Pearson correlation becomes

$$\boxed{C_\theta^{\mathrm{Pea}}(\mathbf{r},t) = \frac{\displaystyle\int_{\mathrm{BZ}} \mathrm{d}^2k\ \cos(\mathbf{k}\cdot\mathbf{r})\ g(\mathbf{k},t)}{\displaystyle\int_{\mathrm{BZ}} \mathrm{d}^2k\ g(\mathbf{k},t)}\,, \qquad g(\mathbf{k},t) \equiv \frac{\Gamma(\mathbf{k})t - 1 + e^{-\Gamma(\mathbf{k})t}}{\Gamma(\mathbf{k})^2},} \tag{3.11}$$

the point $\mathbf{k} = \mathbf{0}$ being excluded: it is of measure zero in the integral, and in the finite-lattice sum it carries the weight $1/N$ that (3.9) sets to zero. This is the quantity that should be compared with the measured shell correlations of Sec. 6.

Writing $\xi = \Gamma(\mathbf{k})t$, the kernel of (3.11) factorises as

$$g(\mathbf{k},t) = \frac{t^2}{2}\cdot\frac{2\left(\xi - 1 + e^{-\xi}\right)}{\xi^2}, \tag{3.12}$$

and the dimensionless factor $\phi(\xi) \equiv 2(\xi - 1 + e^{-\xi})/\xi^2$ has the limits

$$\phi(\xi) \longrightarrow \begin{cases} 1, & \xi \ll 1, \\ \dfrac{2}{\xi} = \dfrac{2}{\Gamma t}, & \xi \gg 1. \end{cases} \tag{3.13}$$

Consequently the single expression (3.11) contains both kernels used in this paper:

- if every mode has saturated, $\Gamma_{\min}t \gg 1$ and hence $\xi \gg 1$ throughout the zone, then $g \to 2t/\Gamma$ and (3.11) reduces to the *stationary* kernel $1/\Gamma$;
- if instead the fast modes have saturated while the slow ones have not, $1 \ll \Gamma_{\mathrm{fast}}t$ and $\Gamma_{\mathrm{slow}}t \ll 1$, then g is t/Γ^2 for the slow modes and $2t/\Gamma$ for the fast ones, so that the integrand is dominated by $1/\Gamma(\mathbf{k})^2$ – the kernel of Eq. (2.10) used in Sec. 2.4.

The two kernels are therefore two windows of (3.11), and no separate assumption is needed to pass between them: the exact kernel interpolates between them as the observation time grows. Since $\Gamma_{\min} = \gamma_0(1-3\mu)$, the crossover between the two windows happens at

$$\Gamma_{\min} t \sim 1 \qquad \Longleftrightarrow \qquad \Gamma(0)t \;\sim\; \frac{1+6\mu}{1-3\mu}, \tag{3.14}$$

so that a larger coupling reaches the stationary window at a smaller $\Gamma(0)t$. This is the sense in which the two kernels address two different physical regimes rather than two approximations to the same regime, and it is why the stationary kernel is the relevant one when the coupling is strong.

3.3 Simulate the stationary kernel $1/\Gamma$

In the limit in which every mode has saturated, (3.11) becomes the central result of this section,

$$\boxed{C_\theta^{\mathrm{Pea}}(\mathbf{r}) = \frac{\int_{\mathrm{BZ}} \mathrm{d}^2k\; e^{i\mathbf{k}\cdot\mathbf{r}}/\Gamma(\mathbf{k})}{\int_{\mathrm{BZ}} \mathrm{d}^2k\; 1/\Gamma(\mathbf{k})}\,, \qquad \Gamma(\mathbf{k}) = \gamma_0 + \gamma S(\mathbf{k}),} \tag{3.15}$$

which depends on the dynamics only through the dimensionless ratio $\mu = \gamma/\gamma_0$, while the noise strength c, the observation time t and the overall scale of Γ all cancel in the normalisation. The correlation is obtained by evaluating the two zone integrals on the same $\mathbf{k}$-grid and taking their ratio. This is legitimate because the integrand is regular on the zone and therefore converges absolutely with no regularisation , and the ratio is well defined for every $\mathbf{r}$. The mode $\mathbf{k} = \mathbf{0}$ is excluded by the definition of the observable with the mean subtraction removes exactly. The one genuine limitation is numerical rather than conceptual in simulation, as $\mu \to 1/3$ the minimum of Γ at the K points becomes sharp, so the quadrature has to resolve a core of width $\sim a^{-1}\sqrt{1-3\mu}$ in $\mathbf{k}$, and for large $|\mathbf{r}|$ it must also resolve $\cos(\mathbf{k}\cdot\mathbf{r})$. Both requirements are met by the scheme of Sec. 3.4. Table 2 compares the stationary kernel with the transient one, both evaluated from the same zone integral.

μ	kernel	shell 1	shell 2	shell 3
$+0.05$	$1/\Gamma^2$	-0.08748	$+0.01265$	$+0.00505$
$+0.05$	$1/\Gamma$	-0.04592	$+0.00440$	$+0.00187$
$+0.10$	$1/\Gamma^2$	-0.15730	$+0.04522$	$+0.01337$
$+0.10$	$1/\Gamma$	-0.08655	$+0.01655$	$+0.00576$
$+0.30$	$1/\Gamma^2$	-0.39504	$+0.50194$	-0.15073
$+0.30$	$1/\Gamma$	-0.26341	$+0.21632$	-0.02775

Table 2: Pearson correlation on the first three shells for the two windows, at the same μ. Both columns are obtained from the same Brillouin-zone integral, with the indicated kernel. The stationary $(1/\Gamma)$ values are uniformly smaller in magnitude than the transient $(1/\Gamma^2)$ ones, and the separation between the two windows grows with μ.

3.4 Numerical evaluation and verification

The sums in (3.11) and (3.15) are evaluated directly, without any further approximation. The integration domain is the hexagonal first Brillouin zone, that is the Voronoi cell of the triangular reciprocal lattice,

$$\mathbf{k} \in \mathrm{BZ} \iff |\mathbf{k}| \le |\mathbf{k} - \mathbf{G}| \quad \text{for every reciprocal vector } \mathbf{G}, \tag{3.16}$$

with the zone average of a function φ defined as $A_{\mathrm{cell}} \int_{\mathrm{BZ}} \mathrm{d}^2k\, \varphi/(2\pi)^2$ and $A_{\mathrm{cell}} = \sqrt{3}a^2/2$. Both kernels $1/\Gamma$ and $1/\Gamma^2$ are largest where Γ is smallest, so the quadrature has to resolve the minimum of Γ at the

K points, a core in $\mathbf{k}$ of width $\sim (\Gamma_{\min}/c_K)^{1/2} \sim a^{-1}\sqrt{1-3\mu}$; the grid must therefore satisfy $\delta k \ll a^{-1}\sqrt{1-3\mu}$, i.e. $\delta k^2 \ll (1-3\mu)/a^2$, as $\mu \to 1/3$. It must also resolve the oscillation $\cos(\mathbf{k}\cdot\mathbf{r})$ for the largest requested r. The zone centre needs no special treatment: it is a smooth maximum of Γ, hence a minimum of the kernel. The result is extrapolated in the grid spacing.

Verification. The evaluation of (3.15) was checked as follows (`lcod_simulations/verify_stationary.py`).

(i) The structure factor obtained from the six nearest-neighbour vectors reproduces the closed form (2.7) to 1.8×10^{-15}, and the zone (3.16) agrees with a brute-force Voronoi test with no disagreements.

(ii) The continuum integral agrees with a direct sum over the allowed wavevectors of a 2048×2048 lattice, with weight $1/N$ each, to 8×10^{-5}, and the shell values change by less than 1.5×10^{-4} between grids of 600 and 4800 points per axis.

(iii) The normalisation $\int_{\mathrm{BZ}} 1/\Gamma$ is finite and positive for $\mu < 1/3$; as $\mu \to 1/3$ it diverges logarithmically, but the numerator is dominated by the same K-point regions, so the ratio remains finite. At fixed μ, the correlation decays to zero at large $|\mathbf{r}|$ with no constant tail. Numerically, the shell-averaged $|C_\theta^{\mathrm{Pea}}|$ is below 10^{-4} beyond the sixth shell at $\mu = 0.04$.

(iv) The program raises if $\Gamma(\mathbf{k}) \le 0$ anywhere, since then the mode is undamped and no stationary correlation exists; with $\Gamma_{\min} = \gamma_0(1-3\mu)$ this occurs for $\mu \ge 1/3$.

The point $\mathbf{k} = \mathbf{0}$ is excluded throughout. In the continuum it is a single point of measure zero, while in a finite sample it would carry the weight $1/N$ and contribute an $\mathbf{r}$-independent term, which is exactly what the mean subtraction (3.9) removes.

3.5 Shell correlations and the decay with distance

Table 3 lists the stationary Pearson correlation (3.15) on the first five shells, and Fig. 5 displays it for several μ. Both the shell structure and the transient counterpart of Sec. 2.4 are non-monotonic, alternating in sign, with the sign of the nearest shell given by the sign of μ. However, its amplitude for a given μ is smaller, because the stationary kernel weights the slow modes less strongly than $1/\Gamma^2$.

μ	Γ_0	shell 1	shell 2	shell 3	shell 4	shell 5
$+0.05$	1.30	-0.04592	$+0.00440$	$+0.00187$	-0.00028	-0.00007
$+0.10$	1.60	-0.08655	$+0.01655$	$+0.00576$	-0.00204	-0.00020
$+0.20$	2.20	-0.16353	$+0.06682$	$+0.01122$	-0.01478	$+0.00329$
$+0.30$	2.80	-0.26341	$+0.21631$	-0.02774	-0.06939	$+0.06202$
$+0.32$	2.92	-0.30284	$+0.31391$	-0.07574	-0.11193	$+0.13553$

Table 3: Pearson correlation C_θ^{Pea} on the first five shells from the stationary kernel (3.15), for a perfect triangular lattice with $a = 1$ and $\gamma_0 = 1$. The values are Pearson-normalised, hence independent of c, t and the overall scale of Γ; they were obtained by extrapolating the Brillouin-zone integral in the grid spacing. The lubrication case is $\mu > 0$, and $\mu < 1/3$ is required for the stationarity of the model.

The decay with distance is shown in Fig. 6. The correlation decays to zero at large $\mathbf{r}$, because the continuous Brillouin-zone integral contains no weight at the single point $\mathbf{k} = \mathbf{0}$, whereas a discrete sum over the allowed wavevectors of a finite sample would assign that mode a weight $1/N$ and hence an $\mathbf{r}$-independent contribution. The envelope of the oscillation is what sets the correlation length, discussed next.

There are several remarks for such result. Firstly the decay length diverges asymptotically as $\mu \to 1/3$. The large-distance decay of (3.15) is controlled by the *minimum* of Γ over the zone and the zone centre

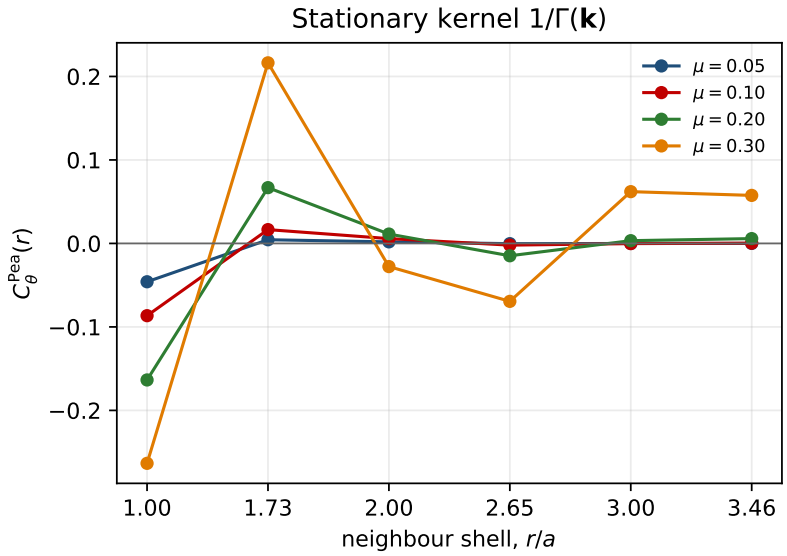


Figure 5: Pearson correlation C_θ^{Pea} on the first six neighbour shells ($r/a = 1, \sqrt{3}, 2, \sqrt{7}, 3, 2\sqrt{3}$) from the stationary kernel (3.15), for several values of $\mu = \gamma/\gamma_0$. The sign of the nearest-neighbour correlation is the sign of μ and the sequence alternates because of the hexagonal anisotropy of $\Gamma(\mathbf{k})$; the amplitude grows with μ because a stronger coupling makes Γ more strongly $\mathbf{k}$-dependent. Compare with Fig. 4, which shows the transient window $1/\Gamma^2$ at the same values of μ.

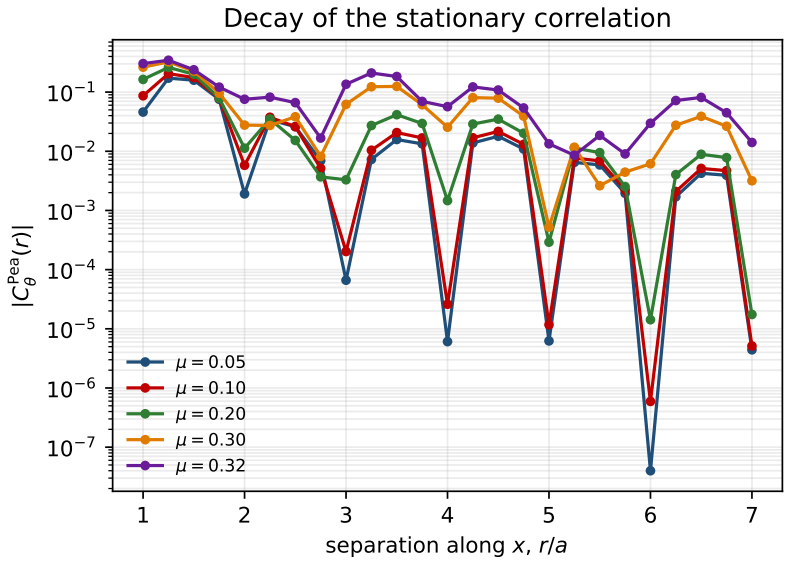


Figure 6: Modulus of the stationary correlation (3.15) along the x axis, for several μ. The oscillation is the hexagonal anisotropy of $\Gamma(\mathbf{k})$; the mean subtraction removes the $\mathbf{k} = \mathbf{0}$ mode, so there is no constant tail.

does not control the decay. What does control it is the K point, where $\Gamma(K) = \gamma_0(1-3\mu)$. Expanding about K,

$$\Gamma(\mathbf{K}+\mathbf{q}) = \Gamma_{\min} + c_K q^2 + \mathcal{O}(q^4), \qquad \Gamma_{\min} = \gamma_0(1-3\mu), \qquad c_K = \tfrac{3}{4}\gamma a^2, \tag{3.17}$$

the kernel is a Lorentzian in $\mathbf{q}$ of width $(\Gamma_{\min}/c_K)^{1/2}$, so the correlation oscillates on the scale $2\pi/|\mathbf{K}|$ and decays with the correlation length (up to an $O(1)$ factor from the 120° wedge of the zone at the corner)

$$\xi_\theta = \left[\frac{c_K}{\Gamma_{\min}}\right]^{1/2} = a\left[\frac{3\mu}{4(1-3\mu)}\right]^{1/2} \xrightarrow[\mu\to 1/3]{} \frac{a}{2\sqrt{1-3\mu}}. \tag{3.18}$$

As $\mu \to 1/3$, ξ_θ diverges as $(1-3\mu)^{-1/2}$ while the normalisation $\int_{\mathrm{BZ}} 1/\Gamma$ diverges logarithmically. The numerator is dominated by the same K-point regions, so the two divergences cancel in the Pearson ratio. The *absolute* variance of $\delta\theta$ grows without bound, but the normalised correlation (3.15) tends to a finite, K-point dominated shape rather than to zero, and its shell amplitudes saturate (the nearest shell approaches -0.4, as the table and Fig. 7 show).

This is quantified in Fig. 7. The measured correlation length from the envelope of $|C_\theta^{\mathrm{Pea}}(r)|$ grows monotonically as $\mu \to 1/3$. Over the range $10^{-2} \le 1-3\mu \le 0.3$ a single effective exponent $\xi_\theta \propto (1-3\mu)^{-p}$ with $p \simeq 0.29$ summarises the data, while the last points ($1-3\mu \simeq 10^{-2}$) approach the asymptotic $p = \frac{1}{2}$ law of (3.18). Reaching the asymptote quantitatively becomes expensive to resolve. The measured length is systematically above the asymptote at large $1-3\mu$, as expected from the admixture of the zone-centre contribution, which is not captured by the K-point expansion alone. The nearest-shell amplitude saturates at about -0.4 in the same range. Since $\Gamma(0)$ grows with μ while $\Gamma(K)$ shrinks, this is also the regime in which the coupling is strong enough to produce a marked shell structure, and it is the regime in which the stationary kernel (3.15), rather than the transient one, is the correct description of a measurement performed over a long window.

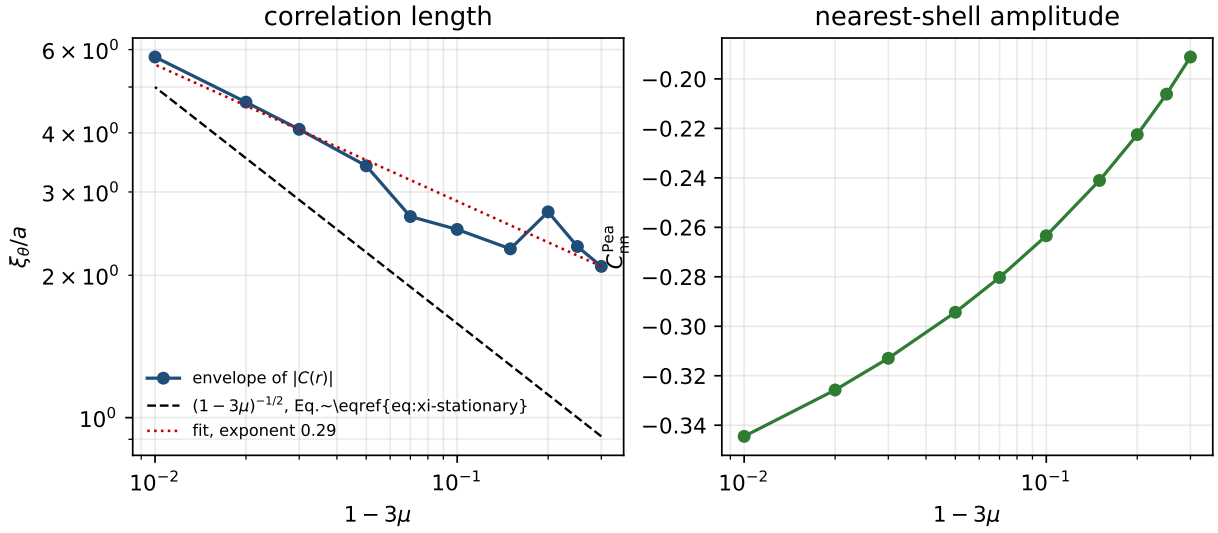


Figure 7: Left: correlation length of the stationary kernel (3.15) as a function of $1-3\mu$, measured from the envelope of $|C_\theta^{\mathrm{Pea}}(r)|$, together with the asymptotic law $(1-3\mu)^{-1/2}$ of (3.18) and the power law fitted to the data. Right: nearest-shell amplitude over the same range; it saturates as the K-point modes soften.

It is worth explaining what changes between the two windows, since the same lattice, the same coupling and the same observable are involved. Both windows are Pearson-normalised, hence independent of c and of the overall scale of Γ, so the comparison is about $\mathbf{k}$ alone. The transient window $1/\Gamma^2$ weights the

soft modes quadratically and therefore emphasises the K-point regions even more strongly; the stationary window $1/\Gamma$ weights them only linearly, so the K points contribute relatively less and the amplitudes are systematically smaller.

4 Generalization to Weakly Disordered Media

The mode analysis of Section 2 considered nearest-neighbour interaction on perfect triangular lattice. In thermally disordered solid or liquid, the discrete set of neighbour vectors is no longer sharp, and the structure factor $S(\mathbf{k})$ of (2.7) is not directly applicable. Nevertheless, the ω-equation in (2.1) is linear and allows simplification. After averaging over *weak disorder*, translational invariance is restored on average, and momentum k remains an approximately good label for the long-wavelength modes, with the discrete neighbour sum replaced by a continuous convolution with an averaged coupling kernel. In this section we make this replacement explicit, derive the disorder-averaged damping, obtain the small-k expansion, and discuss the resulting real-space correlations. The concrete displaced-lattice model, together with its numerical correlations, is treated in Sec. 5.

4.1 From discrete to continuous

For a general disordered medium we replace the discrete neighbour coupling by

$$\sum_{i\in\mathcal{N}(j)} \omega_i(t) \;\longrightarrow\; \int d^2r'\, K(\mathbf{r}_j-\mathbf{r}')\,\omega(\mathbf{r}',t), \tag{4.1}$$

where $K(\mathbf{R})$ is an averaged coupling kernel. In the lattice limit, a sharp set of neighbour vectors is recovered. In a disordered medium we write $R=|\mathbf{R}|, K(\mathbf{R})=g(R)\,J(R)$, where $g(R)$ is the radial distribution function and $J(R)$ is the microscopic coupling kernel. This coupling kernel encodes the angular-velocity increment at one disk induced by that of another, and for interactions in the lubrication limit it is short ranged and peaked about the averaged nearest-neighbour distance.

Using the Fourier transform convention for ω,

$$\omega_{\mathbf{k}}(t)=\frac{1}{\sqrt{\Omega}}\int d^2r\, e^{-i\mathbf{k}\cdot\mathbf{r}}\omega(\mathbf{r},t). \tag{4.2}$$

Strictly speaking, the angular velocities $\omega_j(t)$ are defined only at the discrete particle centres $\mathbf{r}_j$. To discuss long-wavelength modes in a disordered medium, we introduce a coarse-grained angular-velocity field $\omega(\mathbf{r},t)$. This field is not a microscopic replacement for the discrete variables; rather, it is their local average on a scale ℓ satisfying $a\ll\ell\ll k^{-1}$, where a is the typical particle spacing. In this sense, $\omega(\mathbf{r},t)$ represents the slowly varying component of the discrete angular velocities, and its Fourier components coincide with those of the discrete variables in the long-wavelength limit. We use the same symbol ω, but the argument distinguishes the discrete variable $\omega_j(t)$ from the coarse-grained field $\omega(\mathbf{r},t)$. The continuous convolution in (4.1) transforms into $S_{\rm gen}(\mathbf{k})\,\omega_{\mathbf{k}}$, where the generalized structural function is

$$S_{\rm gen}(\mathbf{k})=\int d^2R\, e^{i\mathbf{k}\cdot\mathbf{R}}\,K(\mathbf{R})=\int d^2R\, e^{i\mathbf{k}\cdot\mathbf{R}}\,g(R)J(R). \tag{4.3}$$

In the lattice limit $S_{\rm gen}(\mathbf{k})\longrightarrow S(\mathbf{k})$, with $S(\mathbf{k})$ given by (2.7). Thus (4.3) is the natural continuous generalization of the triangular-lattice structure factor. If disorder fluctuations are neglected, the averaged Fourier mode obeys

$$\frac{d\omega_{\mathbf{k}}}{dt}=-\big[\gamma_0+\gamma S_{\rm gen}(\mathbf{k})\big]\omega_{\mathbf{k}}+c\,\Lambda_{\mathbf{k}}(t). \tag{4.4}$$

Equation (4.4) already shows the main qualitative change, where the sharp lattice structure factor is replaced by the Fourier transform of the averaged pair structure and the coupling kernel.

4.2 Averaged damping with disorder

In a typical weak disordered system, the kernel defined in (4.1) also need modification. In particular, we start from this averaged mode $\bar{K}(\mathbf{R}) = g(R)J(R)$ and include zero mean fluctuation of the coupling kernel

$$\delta K(\mathbf{p}) = \int d^2R\, e^{-i\mathbf{p}\cdot\mathbf{R}}\delta K(\mathbf{R}), \qquad K(\mathbf{R}) = \bar{K}(\mathbf{R}) + \delta K(\mathbf{R}), \qquad \langle \delta K(\mathbf{R})\rangle = 0. \tag{4.5}$$

In Fourier space the mode equation becomes

$$\frac{d\omega_{\mathbf{k}}}{dt} = -\big[\gamma_0 + \gamma S_{\rm gen}(\mathbf{k})\big]\omega_{\mathbf{k}} - \gamma \int \frac{d^2q}{(2\pi)^2}\, \delta K(\mathbf{k}-\mathbf{q})\, \omega_{\mathbf{q}} + c\,\Lambda_{\mathbf{k}}(t), \tag{4.6}$$

We denote the bare, disorder-free damping by $\Gamma_0(\mathbf{k}) = \gamma_0 + \gamma S_{\rm gen}(\mathbf{k})$. To define the propagator, consider first the disorder-free equation (4.4). Let $G_0(\mathbf{k}, t-t')$ be the causal Green function defined by

$$\left(\frac{\partial}{\partial t} + \Gamma_0(\mathbf{k})\right) G_0(\mathbf{k}, t-t') = \delta(t-t'), \qquad G_0(\mathbf{k}, t-t') = 0 \quad \text{for } t < t'. \tag{4.7}$$

In frequency space we use the convention to gain the propagator

$$G_0(\mathbf{k}, z) = \int_0^\infty dt\, e^{izt}\, G_0(\mathbf{k}, t), \qquad \boxed{G_0(\mathbf{k}, z) = \frac{1}{-iz + \Gamma_0(\mathbf{k})}.} \tag{4.8}$$

For the full disordered equation (4.6), we define the exact (discrete) propagator $G_{\rm dis}(\mathbf{k}, \mathbf{q}; z)$ by

$$(-iz + \Gamma_0(\mathbf{k}))G_{\rm dis}(\mathbf{k}, \mathbf{q}; z) + \gamma \int \frac{d^2p}{(2\pi)^2}\, \delta K(\mathbf{k}-\mathbf{p})\, G_{\rm dis}(\mathbf{p}, \mathbf{q}; z) = (2\pi)^2\delta(\mathbf{k}-\mathbf{q}). \tag{4.9}$$

In order to average the propagator over the disordered space, write out the ensemble average of some operator $\mathcal{O}[\delta K]$ over the random coupling fluctuations δK:

$$\langle \mathcal{O}[\delta K]\rangle_{\rm dis} = \int \mathcal{D}[\delta K]\, P[\delta K]\, \mathcal{O}[\delta K],$$

where $P[\delta K]$ is the probability functional (depend on geometry) of the disorder. We assume that the disorder is statistically asymptotically homogeneous (and translation invariant), so that $P[\delta K] \simeq P[T_{\mathbf{a}}\delta K]$ for any translation $T_{\mathbf{a}}$, and the two-point correlation depends only on the separation. Under this assumption, the averaged propagator is diagonal in momentum, hence translation invariance implies

$$\big\langle G_{\rm dis}(\mathbf{k}, \mathbf{q}; z)\big\rangle_{\rm dis} = (2\pi)^2\delta(\mathbf{k}-\mathbf{q})\, G(\mathbf{k}, z), \tag{4.10}$$

where the off-diagonal parts vanish exactly in the thermodynamic limit because they are prohibited by the translational symmetry of the disorder ensemble, which is a direct consequence of statistical homogeneity. In a finite system, non-diagonal terms can appear as finite-size corrections, but they are negligible for the long-wavelength modes considered here.

Such formalism allows us to define the self-energy $\Sigma_{\rm dis}(\mathbf{k}, z)$. The physical meaning is that $\Sigma_{\rm dis}$ collects all irreducible contributions to the averaged propagator by the relation

$$\boxed{G^{-1}(\mathbf{k}, z) = G_0^{-1}(\mathbf{k}, z) + \Sigma_{\rm dis}(\mathbf{k}, z).} \tag{4.11}$$

This result in our convention is a similar form as the Schwinger-Dyson equation for quantum fields. To see this similarity of self energy, iterate (4.11) to obtain the series

$$\begin{aligned} G(\mathbf{k}, z) &= G_0(\mathbf{k}, z) - G_0(\mathbf{k}, z)\Sigma_{\rm dis}(\mathbf{k}, z)G(\mathbf{k}, z) \\ &= G_0(\mathbf{k}, z) - G_0^2(\mathbf{k}, z)\Sigma_{\rm dis}(\mathbf{k}, z) + G_0^2(\mathbf{k}, z)\Sigma_{\rm dis}^2(\mathbf{k}, z)G(\mathbf{k}, z) = \cdots, \end{aligned} \tag{4.12}$$

which shows that indeed $\Sigma_{\rm dis}$ plays the role of the self-energy. In the weak-disorder limit, $\Sigma_{\rm dis}$ can be computed with the same manner in powers of δK, as we do below. To obtain it perturbatively, we start from the exact equation (4.9) and iterate once to write

$$G_{\rm dis}(\mathbf{k},\mathbf{q};z) = G_0(\mathbf{k},z)\,(2\pi)^2\delta(\mathbf{k}-\mathbf{q}) - \gamma\,G_0(\mathbf{k},z)\int\frac{d^2p}{(2\pi)^2}\,\delta K(\mathbf{k}-\mathbf{p})\,G_{\rm dis}(\mathbf{p},\mathbf{q};z).$$

Iterating to second order in δK and then averaging over disorder using $\langle\delta K\rangle_{\rm dis}=0$ and the linear term vanishes, so we obtain

$$\big\langle G_{\rm dis}(\mathbf{k},\mathbf{q};z)\big\rangle_{\rm dis} = (2\pi)^2\delta(\mathbf{k}-\mathbf{q})\,G_0(\mathbf{k},z)\left[1+\gamma^2 G_0(\mathbf{k},z)\int\frac{d^2p}{(2\pi)^2}\,\big\langle|\delta K(\mathbf{k}-\mathbf{p})|^2\big\rangle_{\rm dis}G_0(\mathbf{p},z)\right]+\mathcal{O}(\delta K^4).$$

Here the delta function in the external momenta follows from the statistical homogeneity of the disorder. Comparing (4.2) with (4.12), we identify the self-energy at second order as

$$\Sigma_{\rm dis}(\mathbf{k},z) = -\gamma^2\int\frac{d^2q}{(2\pi)^2}\,\big\langle|\delta K(\mathbf{k}-\mathbf{q})|^2\big\rangle_{\rm dis}\,G_0(\mathbf{q},z)+\mathcal{O}(\delta K^4). \tag{4.13}$$

The minus sign is a consequence of our convention in (4.11), where the self-energy is added to the inverse bare propagator. With either sign of the coupling the second-order self-energy $\Sigma_{\rm dis}=-\gamma^2(\dots)$ is negative, because it is quadratic in γ; i.e. at this order disorder *reduces* the effective damping. This is a genuine (if small) prediction of the weak-disorder expansion.

For the equal-time correlations of the angle field we are interested in times long compared with the microscopic relaxation time Γ_0^{-1}. In this regime the correlation is controlled by the static damping $\Gamma_{\rm gen}(\mathbf{k})$, because the Ornstein–Uhlenbeck process for each mode reaches its stationary variance $c^2/(2\Gamma)$ on time scales $t\gg\Gamma^{-1}$ and the diffusive growth of $\langle|\theta_{\mathbf{k}}|^2\rangle$ is controlled by the zero-frequency response. It therefore suffices to evaluate the self-energy in the low-frequency limit $z\to0$. Using the bare propagator at zero frequency, $G_0(\mathbf{q},0)=\Gamma_0(\mathbf{q})^{-1}$ with $\Gamma_0(\mathbf{q})=\gamma_0+\gamma S_{\rm gen}(\mathbf{q})$, we obtain the static self-energy

$$\boxed{\Sigma_{\rm dis}(\mathbf{k})\equiv\Sigma_{\rm dis}(\mathbf{k},0) = -\gamma^2\int\frac{d^2q}{(2\pi)^2}\,\frac{\big\langle|\delta K(\mathbf{k}-\mathbf{q})|^2\big\rangle_{\rm dis}}{\Gamma_0(\mathbf{q})}+\mathcal{O}(\delta K^4).} \tag{4.14}$$

This static self-energy is the quantity that enters the effective damping and hence the long-time equal-time correlations. The effective damping is therefore

$$\Gamma_{\rm gen}(\mathbf{k}) = \Gamma_0(\mathbf{k})+\Sigma_{\rm dis}(\mathbf{k}) = \gamma_0+\gamma S_{\rm gen}(\mathbf{k})+\Sigma_{\rm dis}(\mathbf{k}). \tag{4.15}$$

This is the disorder-averaged generalization of (2.8). The first two terms are the mean-field and averaged-structure contributions; the self-energy $\Sigma_{\rm dis}(\mathbf{k})$ contains the leading effect of disorder fluctuations. If $\Sigma_{\rm dis}$ is positive, disorder increases the damping; if it is negative, it decreases it.

It is worth noting that (4.14) is written in terms of the zero-frequency propagator $G_0(\mathbf{q},0)=1/\Gamma_0(\mathbf{q})$: the self-energy is therefore already a *stationary* (static) object, and it can be inserted directly into the stationary kernel of Sec. 3. That is precisely what the next subsection does.

This derivation also clarifies the conditions under which (4.15) holds: (i) the disorder is weak, so that the expansion in δK can be truncated at second order; (ii) the disorder is statistically translation invariant, so that the averaged propagator is diagonal in momentum; (iii) the equal-time correlations are considered at times long compared with the microscopic relaxation time, so that the static limit $z\to0$ is appropriate. If the disorder correlation length is comparable to or larger than the wavelength $1/k$, the self-energy can become non-analytic at small k, and the perturbation theory in δK must be reorganised.

4.3 The stationary correlation in a disordered medium

The previous subsection determined the averaged damping (4.15). Since the angle field is diffusive, the disorder-averaged equal-time correlation follows from the same mode decomposition as in Sec. 3, with the bare rate $\Gamma(\mathbf{k})$ replaced by $\Gamma_{\rm gen}(\mathbf{k})$. We state the corresponding stationary covariance first, because it makes the content of that replacement transparent, and then give the finite-time kernel that interpolates between the two windows.

Stationary covariance and the boxed correlation. For each realisation of the disorder, the stationary covariance of the mean-removed angle field follows from the same mode decomposition as in Sec. 3: it is the inverse transform of the zero-frequency response. Averaging over the disorder, statistical homogeneity makes the averaged response diagonal in $\mathbf{k}$, and the Dyson equation (4.11) at $z = 0$ gives

$$G(\mathbf{k}, 0) = \frac{1}{\Gamma_0(\mathbf{k}) + \Sigma_{\rm dis}(\mathbf{k})} = \frac{1}{\Gamma_{\rm gen}(\mathbf{k})}, \tag{4.16}$$

so that the averaged covariance is the inverse transform of $1/\Gamma_{\rm gen}$. Normalising at $\mathbf{r} = \mathbf{0}$ gives the disorder-averaged stationary Pearson correlation

$$\boxed{C^{\rm Pea}_{\rm dis}(\mathbf{r}) = \frac{\displaystyle\int d^2k\ \cos(\mathbf{k}\cdot\mathbf{r})\,/\Gamma_{\rm gen}(\mathbf{k})}{\displaystyle\int d^2k\ 1/\Gamma_{\rm gen}(\mathbf{k})}\ , \qquad \Gamma_{\rm gen} = \gamma_0 + \gamma S_{\rm gen} + \Sigma_{\rm dis},} \tag{4.17}$$

which is the disordered counterpart of (3.15): the only change is the replacement $\Gamma \to \Gamma_{\rm gen}$. The $\mathbf{k} = \mathbf{0}$ subtraction is automatic in the continuum integral, and in a finite sample it is exactly the mean subtraction (3.9), which carries over verbatim from Sec. 3.2. Note that a $\mathbf{k}$-independent part of $\Sigma_{\rm dis}$ cancels between numerator and denominator: the *shape* of the correlation depends only on the $\mathbf{k}$-dependence of the self-energy, and a constant shift of the damping affects neither the shell values nor the correlation length at this order.

Finite observation window. If the self-energy is evaluated at $z = 0$, the effective mode equation is Markovian, and the single-mode result (3.6) of Sec. 3.1 carries over, to leading order in δK, with $\Gamma \to \Gamma_{\rm gen}$:

$$g_{\rm gen}(\mathbf{k}, t) = \frac{\Gamma_{\rm gen}(\mathbf{k})\, t - 1 + e^{-\Gamma_{\rm gen}(\mathbf{k})t}}{\Gamma^2_{\rm gen}(\mathbf{k})}, \tag{4.18}$$

$$\boxed{C^{\rm Pea}_{\rm dis}(\mathbf{r}, t) = \frac{\displaystyle\int d^2k\ \cos(\mathbf{k}\cdot\mathbf{r})\, g_{\rm gen}(\mathbf{k}, t)}{\displaystyle\int d^2k\ g_{\rm gen}(\mathbf{k}, t)}\ .} \tag{4.19}$$

Equation (4.19) contains both disorder results of this paper as windows. In the stationary window, $\Gamma_{\rm gen,min} t \gg 1$ with $\Gamma_{\rm gen,min} = \min_{\mathbf{k}} \Gamma_{\rm gen}(\mathbf{k})$, $g_{\rm gen} \to 2t/\Gamma_{\rm gen}$ and (4.19) reduces to (4.17). In the complementary window in which the fast modes have saturated but the slow ones have not, $g_{\rm gen} \to t/\Gamma^2_{\rm gen}$, and the Pearson correlation is the disordered counterpart of (2.12) with $\Gamma \to \Gamma_{\rm gen}$; this is the kernel used for the displaced-lattice results of Sec. 5.2. In the third regime, in which no mode has saturated ($\Gamma_{\rm gen,max} t \ll 1$), $g_{\rm gen} \to t^2/2$ is $\mathbf{k}$-independent, so the normalised correlation degenerates to a delta function and no spatial structure is measurable; this is the disordered analogue of the ballistic window of Sec. 3.1.

The static approximation $\Sigma_{\rm dis}(\mathbf{k}, z) \simeq \Sigma_{\rm dis}(\mathbf{k}, 0)$ requires the memory time of the disorder to be short compared with the observation window. If it is not, $g_{\rm gen}$ must be replaced by the solution of the corresponding memory equation; this is a correction of order δK^2 and does not affect the qualitative conclusions below.

Special cases and consistency. For $\delta K = 0$, that is, when the disorder is represented only by the averaged kernel $\bar{K}$ (the displaced-lattice model of Sec. 5.1), the self-energy vanishes, $\Gamma_{\mathrm{gen}} = \Gamma_{\mathrm{dis}}$ of (5.8), and (4.17) is the stationary correlation of the displaced lattice; as $\sigma \to 0$ it reduces to (3.15), which is the consistency check with Sec. 3. For $\delta K \neq 0$ the self-energy (4.14) is negative at second order, $\Sigma_{\mathrm{dis}} < 0$, so that disorder *reduces* the effective damping at this order; in (4.17) only its $\mathbf{k}$-dependence survives, so the leading effect on the shell structure is the curvature of the self-energy rather than its value. Because S_{dis} and Σ_{dis} enter Γ_{gen} additively at this order, the two disorder effects—the change of the average structure factor and the fluctuations of the kernel about it—can be discussed separately.

4.4 Correlation length and real-space asymptotics

The real-space behaviour of (4.17) is the Fourier transform of $1/\Gamma_{\mathrm{gen}}$, normalised at the origin. Since $1/\Gamma_{\mathrm{gen}}$ is largest where Γ_{gen} is smallest, the large-distance decay is controlled by the *minimum* of the damping over the zone, and not by its value at $\mathbf{k} = \mathbf{0}$. This point deserves emphasis, because for the ordered lubrication lattice $\Gamma(\mathbf{k})$ has a *maximum* at $\mathbf{k} = \mathbf{0}$ ($S(\mathbf{0}) = 6$) and its minimum at the K points; disorder, by changing $S_{\mathrm{gen}}(\mathbf{0})$ and the curvature of S_{gen}, can move the minimum and hence change which wavevectors control the decay.

Minimum at the zone centre. Suppose first that Γ_{gen} has its minimum at $\mathbf{k} = \mathbf{0}$, as happens when $S_{\mathrm{gen}}(\mathbf{0})$ is the smallest value of S_{gen}. Writing

$$S_{\mathrm{gen}}(\mathbf{k}) = S_{\mathrm{gen}}(\mathbf{0}) + \tfrac{1}{2} S_2 k^2 + \mathcal{O}(k^4), \qquad \Sigma_{\mathrm{dis}}(\mathbf{k}) = \Sigma_0 + \tfrac{1}{2}\Sigma_2 k^2 + \mathcal{O}(k^4), \tag{4.20}$$

with $\Sigma_0 = \Sigma_{\mathrm{dis}}(\mathbf{0})$, one has

$$\Gamma_{\mathrm{gen}}(\mathbf{k}) = \Gamma_{\mathrm{gen}}(\mathbf{0})\big[1 + \kappa k^2\big] + \mathcal{O}(k^4), \qquad \kappa = \frac{\gamma S_2 + \Sigma_2}{2\,\Gamma_{\mathrm{gen}}(\mathbf{0})}, \tag{4.21}$$

which is a minimum provided $\gamma S_2 + \Sigma_2 > 0$. Then $1/\Gamma_{\mathrm{gen}} \simeq \Gamma_{\mathrm{gen}}(\mathbf{0})^{-1}(1+\kappa k^2)^{-1}$, whose two-dimensional Fourier transform is a screened Coulomb profile $K_0(r/\xi_\theta)$, so that the correlation is monotonic at large r with the correlation length

$$\xi_\theta = \frac{1}{\sqrt{\kappa}} = \left[\frac{2\,\Gamma_{\mathrm{gen}}(\mathbf{0})}{\gamma S_2 + \Sigma_2}\right]^{1/2}. \tag{4.22}$$

For the ordered lattice $S_2 = -3a^2 < 0$, so this case does not apply: the minimum is not at the zone centre, and the decay is controlled by the wavevector at which Γ_{gen} does attain its minimum. The displaced-lattice model of Sec. 5.1 also keeps $S_{\mathrm{dis}}(\mathbf{0}) = 6$, so there the maximum remains at the zone centre and only its height is reduced by the Debye–Waller factor; case (A) is therefore the one appropriate to a liquid-like structure factor with a small $S_{\mathrm{gen}}(\mathbf{0})$, rather than to a weakly disordered solid.

Minimum at a finite wavevector, and the approach to criticality. For the ordered lubrication lattice, and more generally whenever S_{gen} has its minimum at a finite wavevector, the minimum of Γ_{gen} sits at the K points (or on a ring of equivalent points). Expanding about the minimum, $\Gamma_{\mathrm{gen}}(\mathbf{k}_0 + \mathbf{q}) = \Gamma_{\mathrm{gen}}(\mathbf{k}_0) + c_0 q^2 + \mathcal{O}(q^4)$, the kernel is a Lorentzian in $\mathbf{q}$ of width $(\Gamma_{\mathrm{gen}}(\mathbf{k}_0)/c_0)^{1/2}$, so that the correlation oscillates with wavevector $\mathbf{k}_0$ and decays with the length

$$\xi_\theta = \left[\frac{c_0}{\Gamma_{\mathrm{gen}}(\mathbf{k}_0)}\right]^{1/2}, \qquad C^{\mathrm{Pea}}_{\mathrm{dis}}(\mathbf{r}) \sim A\,\frac{e^{-r/\xi_\theta}}{\sqrt{r}}\,\cos(\mathbf{k}_0\cdot\mathbf{r} + \phi), \qquad A \sim \frac{1}{c_0 I_0}, \tag{4.23}$$

where $I_0 = \int d^2k\, 1/\Gamma_{\rm gen}(\mathbf{k})$ and ϕ is a phase fixed by the local anisotropy. This is the smooth, possibly oscillatory, short-ranged correlation that replaces the discrete shell structure of the ordered lattice, and it is the form to be compared with the smooth correlations measured in a disordered medium.

As the minimum softens, $\Gamma_{\rm gen}(\mathbf{k}_0) \to 0$, the length (4.23) diverges while the normalisation $\int d^2k\, 1/\Gamma_{\rm gen}$ diverges as well. The numerator is dominated by the same neighbourhood of $\mathbf{k}_0$, so the divergences cancel in the normalised ratio: the absolute variance grows without bound, but the correlation (4.17) tends to a finite, $\mathbf{k}_0$-dominated shape. This is the disorder counterpart of the K-point softening discussed in Sec. 3.5, and it identifies the $1/\sqrt{1-3\mu}$ law found there as a K-point law. Indeed, for the ordered lattice $c_0 = \frac{3}{4}\gamma a^2$ and $\Gamma(K) = \gamma_0(1-3\mu)$, so that

$$\xi_\theta \simeq a\left[\frac{3\mu}{4\,(1-3\mu)}\right]^{1/2} \xrightarrow[\mu\to 1/3]{} \frac{a}{2\sqrt{1-3\mu}}, \tag{4.24}$$

and disorder can only shift the value of μ at which the minimum vanishes, through $\Sigma_{\rm dis}$.

Relation to the two windows. The finite-time expression (4.19) has the same geometric content: the window selects how strongly the soft modes are weighted, but the *range* of the correlation is always set by the minimum of $\Gamma_{\rm gen}$, and its *oscillation* by the position of that minimum. The stationary window weights the minimum linearly and the transient window quadratically, so the transient correlation is more strongly dominated by the softest region and therefore has a longer effective range for the same $\Gamma_{\rm gen}$. This is the disordered analogue of the comparison (3.15)–(2.12) in the ordered case.

4.5 Experimental determination of $S_{\rm gen}(\mathbf{k})$

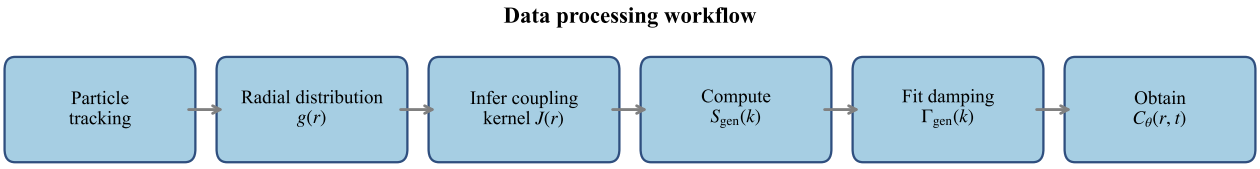


Figure 8: Data processing workflow for extracting the generalized structure factor $S_{\rm gen}(k)$ and the effective damping $\Gamma_{\rm gen}(k)$ from experimental particle tracking data. The workflow proceeds from particle tracking to the radial distribution function $g(r)$, inference of the coupling kernel $J(r)$, computation of $S_{\rm gen}(k)$, fitting of $\Gamma_{\rm gen}(k)$, and finally obtaining the equal-time correlation $C_\theta(r,t)$.

The generalized structural function $S_{\rm gen}(\mathbf{k})$ provides the bridge between the microscopic structure of the disordered medium and the effective damping $\Gamma_{\rm gen}(\mathbf{k})$. It can be determined experimentally as follows.

1. **Measure the static structure.** From particle tracking, compute the radial distribution function $g(r)$ of the particle centres. In two dimensions this is obtained by histogramming the pair distances and normalizing by the ideal-gas expectation at the same density. The result should be checked for finite-size effects by comparing different system sizes.

2. **Infer the coupling kernel.** The kernel $J(r)$ may be known from the microscopic model, for example from lubrication theory, hydrodynamic interactions, or direct force measurements. If it is not known

independently, it can be calibrated by comparing measured mode relaxation rates with (4.15). Alternatively, one can measure the dynamic structure factor of ω or θ, extract $\Gamma_{\rm gen}(\mathbf{k})$ by fitting a Lorentzian in frequency, and then invert

$$\Gamma_{\rm gen}(\mathbf{k}) = \gamma_0 + \gamma S_{\rm gen}(\mathbf{k}) + \Sigma_{\rm dis}(\mathbf{k}) \tag{4.25}$$

to obtain $S_{\rm gen}(\mathbf{k})$, provided γ_0 and $\Sigma_{\rm dis}(\mathbf{k})$ are separately estimated or varied controllably.

3. **Compute the Fourier transform.** Once $g(r)J(r)$ is known, evaluate

$$S_{\rm gen}(\mathbf{k}) = \int d^2r\, e^{i\mathbf{k}\cdot\mathbf{r}}\, g(r)J(r) \tag{4.26}$$

by numerical integration. In an isotropic medium, angular averaging gives $S_{\rm gen}(k)$. The result is then used directly in (4.15) to obtain the effective damping.

4. **Form the correlation.** With $\Gamma_{\rm gen}(\mathbf{k})$ fixed, the stationary correlation follows from (4.17) and the finite-window correlation from (4.19); the window actually realised is selected by $\Gamma_{\rm gen,min}t$, exactly as in the ordered case (Sec. 3.1). Since the self-energy is negative and of order δK^2, measurements at two disorder strengths suffice to separate the averaged-structure contribution $S_{\rm gen}$ from the fluctuation contribution $\Sigma_{\rm dis}$, provided the same $J(r)$ is used in both.

In summary, weak disorder does not destroy the mode description. It replaces the sharp lattice structure factor by the smooth generalized structure factor $S_{\rm gen}(\mathbf{k})$, renormalizes the damping through the disorder self-energy $\Sigma_{\rm dis}(\mathbf{k})$, and turns the discrete shell correlations into smooth, possibly oscillatory, short-range correlations. The entire content of Sec. 3—the boxed stationary correlation (3.15), its two windows, and its decay length—carries over to the disordered medium through the single substitution $\Gamma(\mathbf{k}) \to \Gamma_{\rm gen}(\mathbf{k})$, as stated in (4.17) and (4.19). The mean-field result is recovered only when $S_{\rm gen}(\mathbf{k})$ is replaced by its $\mathbf{k} = 0$ value and all disorder self-energy corrections are neglected.

5 Lattice disorder: the displaced lattice and its correlations

Section 4 kept the coupling kernel general. We now make the disorder average explicit with the displaced-lattice model, for which it can be carried out in closed form, and evaluate the two correlations of Secs. 2.4 and 3 with the disordered damping. The only change to the coupling is linear in γ: the Bragg part of the structure factor is reduced by the Debye–Waller factor $|\phi|^2$ and a constant background $6(1-|\phi|^2)$ is added, so that the Brillouin-zone machinery of the two preceding sections applies without modification.

5.1 Disorder-averaged structure factor of a displaced lattice

A concrete realisation of the averaged kernel, and the one used for the numerical results below, is a lattice in which every particle centre carries an independent random displacement $\mathbf{u}_j$,

$$\mathbf{r}_j = \mathbf{R}_j + \mathbf{u}_j, \qquad \langle \mathbf{u}_j \rangle = 0, \qquad \langle |\mathbf{u}_j|^2 \rangle = \sigma^2, \qquad \mathbf{u}_j \text{ i.i.d.} \tag{5.1}$$

with $\mathbf{R}_j$ the sites of the perfect triangular lattice. Two displacement statistics are of interest.

(i) Isotropic Gaussian displacements.

$$P(\mathbf{u}) = \frac{1}{2\pi s^2} e^{-|\mathbf{u}|^2/(2s^2)}, \qquad \sigma^2 = \langle |\mathbf{u}|^2 \rangle = 2s^2, \qquad \phi(\mathbf{k}) = \left\langle e^{i\mathbf{k}\cdot\mathbf{u}} \right\rangle = e^{-\sigma^2 k^2/4}. \tag{5.2}$$

(ii) Fixed displacement magnitude, random direction.

$$P(\mathbf{u}) = \frac{1}{2\pi\sigma}\delta(|\mathbf{u}| - \sigma), \qquad \phi(\mathbf{k}) = \left\langle e^{i\mathbf{k}\cdot\mathbf{u}} \right\rangle = J_0(k\sigma). \tag{5.3}$$

Both characteristic functions share the small-k behaviour

$$\phi(\mathbf{k}) = 1 - \frac{\sigma^2 k^2}{4} + \mathcal{O}(k^4), \tag{5.4}$$

so the two displacement statistics are indistinguishable for $k\sigma \ll 1$, and they differ only when the wavelength becomes comparable to the displacement itself. In the code the parameter `sigma` always denotes the rms magnitude $\sqrt{\langle|\mathbf{u}|^2\rangle}$ of the displacement, consistent with (5.2)–(5.3).

The structure factor of the displaced lattice is the ensemble average

$$S(\mathbf{k}) = \frac{1}{N}\Big\langle \Big|\sum_j e^{i\mathbf{k}\cdot\mathbf{r}_j}\Big|^2 \Big\rangle = \frac{1}{N}\sum_{j,l} e^{i\mathbf{k}\cdot(\mathbf{R}_j - \mathbf{R}_l)} \big\langle e^{i\mathbf{k}\cdot(\mathbf{u}_j - \mathbf{u}_l)} \big\rangle. \tag{5.5}$$

For $j \neq l$ the displacements are independent, so the average factorises into $|\phi(\mathbf{k})|^2$; for $j = l$ it equals one. Hence

$$S_{\rm dis}(\mathbf{k}) = \big(1 - |\phi(\mathbf{k})|^2\big) + \frac{|\phi(\mathbf{k})|^2}{N}\Big|\sum_j e^{i\mathbf{k}\cdot\mathbf{R}_j}\Big|^2 = \big(1 - |\phi|^2\big) + |\phi|^2\,\frac{S_{\rm perfect}(\mathbf{k})}{N}, \tag{5.6}$$

the standard Debye–Waller decomposition: a diffuse background of weight $1 - |\phi|^2$ per particle plus the Bragg peaks suppressed by $|\phi|^2$.

For the damping it is the *per-particle* normalisation that must be used, because $\Gamma(\mathbf{k}) = \gamma_0 + \gamma S(\mathbf{k})$ is built from the six-cosine sum with $S(0) = 6$, i.e. from a quantity that stays finite as $N \to \infty$ whereas (5.6) is extensive in the Bragg term. Rescaling (5.6) so that the Bragg part carries the same weight $S_{\rm perfect}(0) = 6$ gives the form used throughout this work,

$$\boxed{S_{\rm dis}(\mathbf{k}) = |\phi(\mathbf{k})|^2\, S_{\rm perfect}(\mathbf{k}) + 6\big(1 - |\phi(\mathbf{k})|^2\big)\ ,} \tag{5.7}$$

which has three properties that the raw form lacks: (i) $S_{\rm dis}(\mathbf{k}) \to S_{\rm perfect}(\mathbf{k})$ exactly as $\sigma \to 0$, so the perfect lattice is recovered without any residual background; (ii) $S_{\rm dis}(\mathbf{0}) = 6|\phi(\mathbf{0})|^2 = 6$, so $\Gamma_{\rm dis}(\mathbf{0}) = \Gamma_0$ is independent of the disorder strength, and hence of the number of particles; (iii) $S_{\rm dis}$ is bounded, $|S_{\rm dis}| < 6$, so $\Gamma_{\rm dis}$ stays positive for the same range of μ as in the ordered case. Property (iii) is what allows the disorder to be switched on continuously in the numerical scheme; the extensive form (5.6) violates it because its background grows with the sample.

With (5.7) the damping becomes

$$\Gamma_{\rm dis}(\mathbf{k}) = \gamma_0 + \gamma S_{\rm dis}(\mathbf{k}) = \gamma_0 + \gamma|\phi|^2 S_{\rm perfect}(\mathbf{k}) + 6\gamma\big(1 - |\phi|^2\big), \tag{5.8}$$

and the Pearson shell correlation is again obtained from the exact sum (2.12) with $\Gamma \to \Gamma_{\rm dis}$.

5.2 Numerical correlations of the displaced lattice

With $\Gamma_{\rm dis}$ of (5.8), the two windows of Secs. 2.4 and 3 become

$$C^{\rm Pea}_{{\rm dis},2}(\mathbf{r}) = \frac{\int_{\rm BZ} \mathrm{d}^2k\, \cos(\mathbf{k}\cdot\mathbf{r})/\Gamma^2_{\rm dis}}{\int_{\rm BZ} \mathrm{d}^2k\, 1/\Gamma^2_{\rm dis}}, \qquad C^{\rm Pea}_{{\rm dis},1}(\mathbf{r}) = \frac{\int_{\rm BZ} \mathrm{d}^2k\, \cos(\mathbf{k}\cdot\mathbf{r})/\Gamma_{\rm dis}}{\int_{\rm BZ} \mathrm{d}^2k\, 1/\Gamma_{\rm dis}}, \tag{5.9}$$

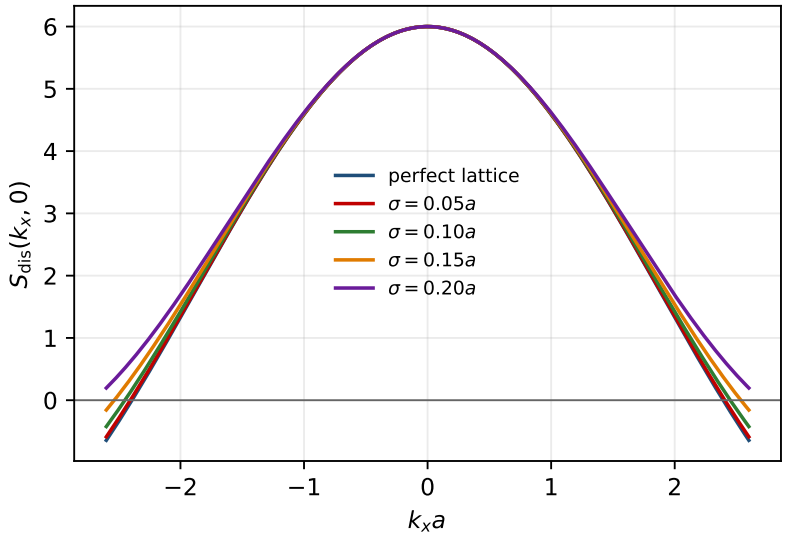


Figure 9: Disorder-averaged structure factor $S_{\mathrm{dis}}(k_x, 0)$ of (5.7) for the perfect lattice and for four displacement strengths. The Debye–Waller factor suppresses the zone-centre peak and lifts the minima, reducing the anisotropy of the coupling while keeping $S_{\mathrm{dis}}(\mathbf{0}) = 6$.

where the subscript 2 denotes the transient window and 1 the stationary one. The disorder enters only through

$$\Gamma_{\mathrm{dis}}(\mathbf{k}) = \Gamma_0 + \gamma |\phi(\mathbf{k})|^2 \big[S_{\mathrm{perfect}}(\mathbf{k}) - 6 \big], \tag{5.10}$$

a linear correction to the coupling: the Bragg part is scaled by $|\phi|^2$, and the constant $6\gamma(1-|\phi|^2)$ is absorbed into Γ_0. Equations (5.9) are evaluated with the same Brillouin-zone quadrature as in Sec. 3.4; no further approximation is made.

Table 4 lists the transient-window values and Table 5 the stationary-window values, both for $\mu = 0.05$ and Gaussian displacements. Disorder reduces the magnitude of every shell and flattens the shell-to-shell contrast in both windows. In the transient window the nearest shell falls from -0.0875 to -0.0617 between $\sigma = 0$ and $0.20a$, while the second shell is suppressed much more strongly, from $+0.0127$ to -0.0027, i.e. it changes sign. The stationary window behaves in the same way, with the nearest shell going from -0.0459 to -0.0322 and the second from $+0.0044$ to -0.0023. The relative suppression of the nearest shell is almost identical in the two windows, a factor 0.70, so the two kernels respond to displacement disorder in the same way even though their absolute amplitudes differ by about a factor two. Beyond $\sigma \simeq 0.15a$ the alternating tail has essentially disappeared. This is the quantitative version of the statement that a disordered lattice has a smooth, short-ranged correlation rather than a sharp shell structure.

The fixed-magnitude, random-direction statistic (5.3) agrees with the Gaussian one to better than 1% for $\sigma \le 0.05a$. At larger displacements the two differ at the level of the shell amplitudes: for the transient window the largest relative differences over the shells shown are about 6%, 8% and 22% at $\sigma = 0.10a$, $0.15a$ and $0.20a$, the last on the second shell, which is close to its sign change; for the stationary window the differences are smaller. The shape of the displacement distribution therefore matters once σ reaches a sizeable fraction of the lattice constant, as expected from (5.4): the two ϕ's differ at order $k^4\sigma^4$ while the shells of interest involve $k\sigma$ of order unity.

The evaluation was checked as follows. (i) $S_{\mathrm{dis}}(\mathbf{0}) = 6$ for every σ, so $\Gamma_{\mathrm{dis}}(\mathbf{0}) = \Gamma_0$ is independent of the disorder strength. (ii) As $\sigma \to 0$ the two columns reduce exactly to the ordered tables of Secs. 2.4 and 3; in the numerical evaluation the difference is zero to machine precision. (iii) $\Gamma_{\mathrm{dis}}(\mathbf{k}) > 0$ for every

σ/a	shell 1	shell 2	shell 3	shell 4	shell 5
0 (perfect)	-0.08750	$+0.01266$	$+0.00506$	-0.00111	-0.00021
0.05	-0.08560	$+0.01139$	$+0.00472$	-0.00072	-0.00025
0.10	-0.08018	$+0.00787$	$+0.00371$	$+0.00034$	-0.00032
0.15	-0.07190	$+0.00288$	$+0.00210$	$+0.00174$	-0.00031
0.20	-0.06169	-0.00265	$+0.00002$	$+0.00313$	-0.00018

Table 4: Transient-window Pearson correlation $C^{\mathrm{Pea}}_{\mathrm{dis},2}$ of (5.9) on the first five shells, for $\mu = 0.05$ and Gaussian displacements, from the Brillouin-zone integral with Γ_{dis} of (5.8). The values are Pearson-normalised and were obtained by extrapolating the integral in the grid spacing. Disorder reduces the magnitude of every shell and flattens the shell-to-shell contrast, removing the alternating tail first.

σ/a	shell 1	shell 2	shell 3	shell 4	shell 5
0 (perfect)	-0.04593	$+0.00443$	$+0.00188$	-0.00029	-0.00007
0.05	-0.04492	$+0.00387$	$+0.00171$	-0.00013	-0.00008
0.10	-0.04204	$+0.00233$	$+0.00121$	$+0.00031$	-0.00009
0.15	-0.03764	$+0.00013$	$+0.00042$	$+0.00090$	-0.00009
0.20	-0.03223	-0.00233	-0.00056	$+0.00149$	-0.00004

Table 5: Stationary-window Pearson correlation $C^{\mathrm{Pea}}_{\mathrm{dis},1}$ of (5.9) on the first five shells, under the same conditions as Table 4. The stationary values are about half the transient ones at $\sigma = 0$, and their relative suppression by disorder is the same.

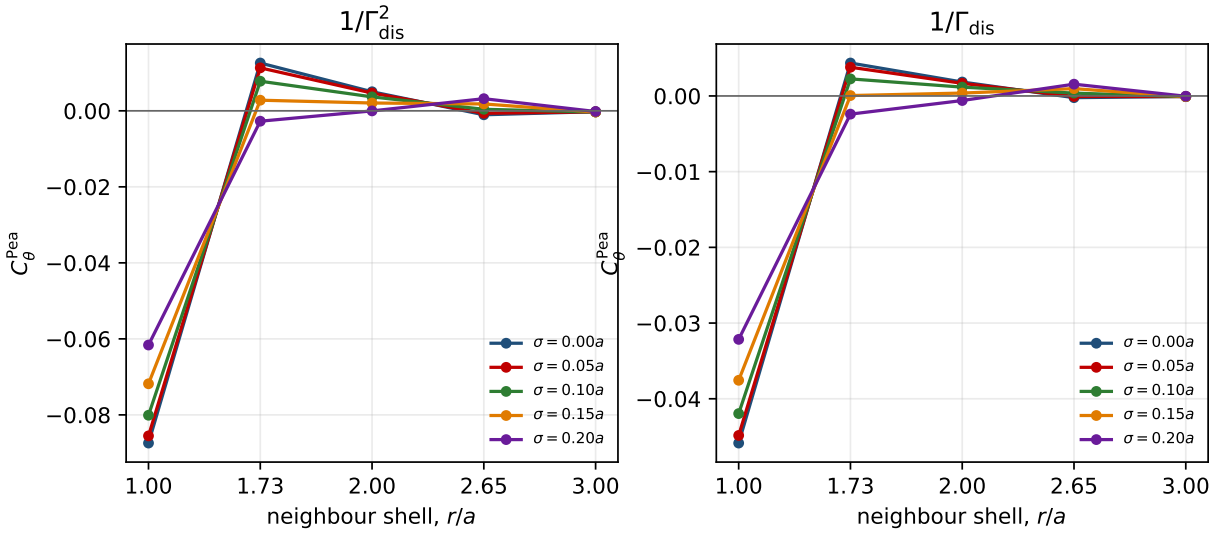


Figure 10: Pearson correlation on the first five shells for $\mu = 0.05$ and increasing displacement disorder, in the transient window $1/\Gamma^2_{\mathrm{dis}}$ (left) and the stationary window $1/\Gamma_{\mathrm{dis}}$ (right). Both windows show the same qualitative behaviour: the magnitude decreases and the shell-to-shell contrast is flattened, with the second shell changing sign between $\sigma = 0.15a$ and $0.20a$.

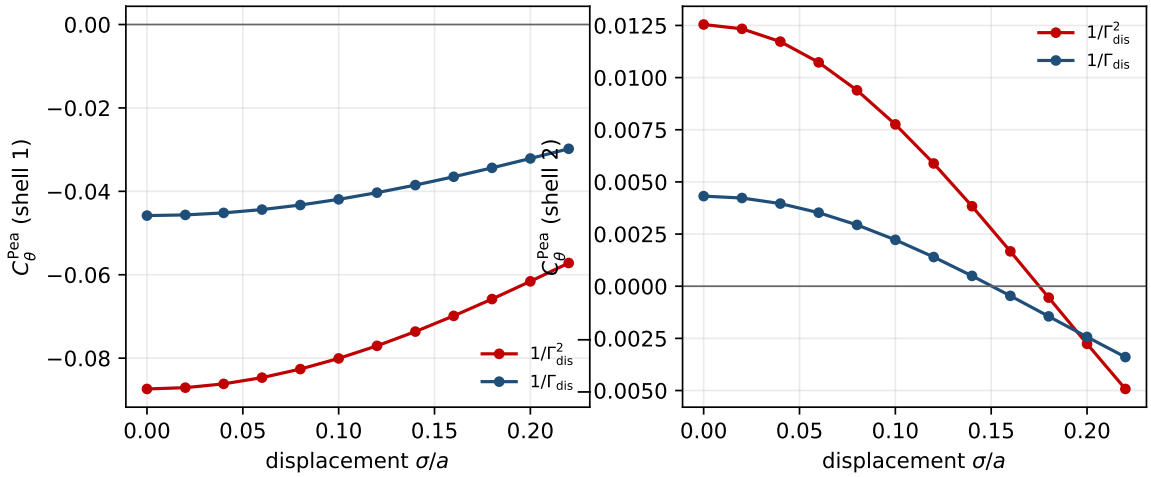


Figure 11: Nearest-shell (left) and second-shell (right) correlation as a function of the displacement strength σ/a, for the two windows at $\mu = 0.05$. Each curve passes through zero where the corresponding shell changes sign; the stationary window is roughly a factor two smaller in amplitude but has the same shape.

$\mu < 1/3$ and every σ shown, so both kernels remain integrable. (iv) At $\sigma = 0.20a$ the shell values change by less than 3×10^{-4} between grids of 1200 and 4800 points per axis.

6 Experimental Observations

The theory developed above predicts a shell-resolved, equal-time correlation of the orientational field on the triangular lattice (Eq. (2.12) and Table 1). We now report an experimental measurement of this quantity.

The experiment consists of a two-dimensional assembly of optically driven, rotating disks on a triangular lattice, imaged and particle-tracked over [N_FRAMES] frames ($\Delta t =$ [DT] s per frame; disk diameter [DIAM] μm). In the present data set 1199 frames were analysed, containing about 390 tracked disks per frame. To avoid defects and boundary effects, the analysis is restricted to the hexagonal regions of the assembly: a particle is retained only if its Voronoi cell (the dual of the Delaunay triangulation) has exactly six neighbours, which leaves about 283 particles per frame. Neighbours of a retained particle are grouped into distance shells of the triangular lattice, labelled by $r/a = 1, \sqrt{3}, 2, \sqrt{7}, \ldots$ with a the nearest-neighbour spacing.

The quantity we extract is the equal-time Pearson correlation of the orientational field,

$$C_\theta^{\mathrm{Pea}}(\mathbf{r}, 0) \;=\; \frac{\big\langle \delta\theta_{\mathbf{r}}(0)\, \delta\theta_{\mathbf{0}}(0) \big\rangle}{\sqrt{\big\langle \delta\theta_{\mathbf{r}}(0)^2 \big\rangle \, \big\langle \delta\theta_{\mathbf{0}}(0)^2 \big\rangle}}, \qquad \delta\theta \equiv \theta - \langle\theta\rangle_{\mathrm{frame}}, \tag{6.1}$$

where the average runs over all retained reference particles and over all frames, and the frame-wise orientational mean $\langle\theta\rangle_{\mathrm{frame}}$ is subtracted because the whole field slowly reorients during the acquisition. The estimator (6.1) is dimensionless, bounded by unity, and independent of the arbitrary zero of the angular scale; it is the experimentally most faithful realisation of the shell-resolved correlation computed in Sec. 2.4. Statistical significance is assessed with a null model in which the orientational values are reshuffled among the retained particles of each frame (destroying any spatial correlation while preserving the geometry, the angular distribution and the frame structure); 200 such realisations on 400 frames give the null distribution quoted below.

Figure 12 shows C_θ^{Pea} on the first four shells, together with the reshuffled-identity null ($\pm 3\sigma$). The measured values are small in magnitude, $C_\theta^{\mathrm{Pea}} = [+0.0033] \pm [0.0029]$ on the first shell, -0.0218 ± 0.0030 on the second, -0.0119 ± 0.0031 on the third and -0.0095 ± 0.0029 on the fourth (errors are frame-to-frame standard errors of the mean). Such small values are expected: the coupling we probe is a noise-mediated lubrication coupling, and in a strongly noisy environment the induced correlations must be weak. Smallness, however, does not imply a null result. The second and third shells lie 9.5σ and 3.8σ away from the shuffled-identity null (deviations with $p < 0.001$ and $p = 0.005$, respectively), while the first shell ($p = 0.67$) and the fourth shell ($p = 0.055$) are consistent with the null. The correlation on the second and third shells is therefore established at a highly significant level despite its small absolute magnitude, whereas the signal on the remaining shells is below the present noise level.

Because the rotational noise is strong, the coupling is weak relative to it, and no coherent, deterministic time-evolution rule could be extracted from the data: the time-dependent mode structure predicted in Secs. 2–4 remains untested experimentally. A quantitative fit of γ_0, γ and c to the measured C_θ^{Pea} is likewise left for future work with lower-noise data.

It is nevertheless worth comparing the measurement with the prediction of Sec. 2.4. The measured pattern, $\big(+0.003,\ -0.022,\ -0.012,\ -0.010\big)$ on shells 1–4, is essentially zero on the nearest shell and *negative* on the next three, with the largest magnitude on the second shell. The lubrication coupling of Sec. 1 corresponds to $\mu > 0$, and Table 1 then gives a *negative* nearest shell with an alternating tail of rapidly decreasing magnitude; the overall sign is therefore consistent, and a least-squares comparison over shells 1–4 is minimised at $\mu \simeq 2 \times 10^{-3}$, i.e. a coupling some two orders of magnitude weaker than the bulk damping, consistent with the conclusion drawn from the smallness of the measured values.

What the minimal model does *not* reproduce is the measured ratio: the data give $C_2/C_1 \simeq -6.6$, whereas the nearest-neighbour-only structure factor gives $|C_2/C_1| \lesssim 0.3$ for every μ (it approaches 0 as $\mu \to 0$). The discrepancy is not a failure of the numerical scheme – the integral (2.11) was verified against a direct real-space inversion of the drift matrix and against the finite-lattice sum – but a statement about the model: a coupling that is strictly nearest-neighbour and isotropic in $\mathbf{k}$ cannot produce a correlation that is an order of magnitude larger on the second shell than on the first. Reproducing the measured ratios requires one of

- a longer-ranged or angle-dependent coupling kernel $K(\mathbf{R})$, i.e. the generalised structure factor $S_{\mathrm{gen}}(\mathbf{k})$ of Sec. 4, whose additional Fourier components weight the outer shells differently;
- a finite experimental resolution in $\mathbf{k}$: the measurement is a window average, and the resolution kernel suppresses exactly the $\mathbf{k} \to 0$ region which, in the model, carries most of the nearest-shell signal;
- a coupling model in which several rotational degrees of freedom per disk (or the finite disk deformation) contribute, which again enriches the $\mathbf{k}$-dependence of the damping.

We flag this as the main quantitative open point of the comparison. It is notable that the measured magnitudes are all very small, $|C_\theta^{\mathrm{Pea}}| \lesssim 0.02$, which is itself consistent with a coupling much weaker than the bulk damping.

The values reported above are given in the convention of Eq. (6.1); as usual for an orientational correlation, the overall sign depends on the convention adopted for the angular variable, which should be matched when comparing with Eq. (2.12).

shell	r/a	C_θ^{Pea}	error	p (shuffled null)
1	1.000	$+0.0033$	0.0029	0.67
2	1.732	-0.0218	0.0030	< 0.001
3	2.000	-0.0119	0.0031	0.005
4	2.646	-0.0095	0.0029	0.055

Table 6: Equal-time Pearson correlation of the orientational field on the first four neighbour shells, with frame-to-frame standard errors and the p-values obtained from 200 reshuffled-identity realisations (see text).

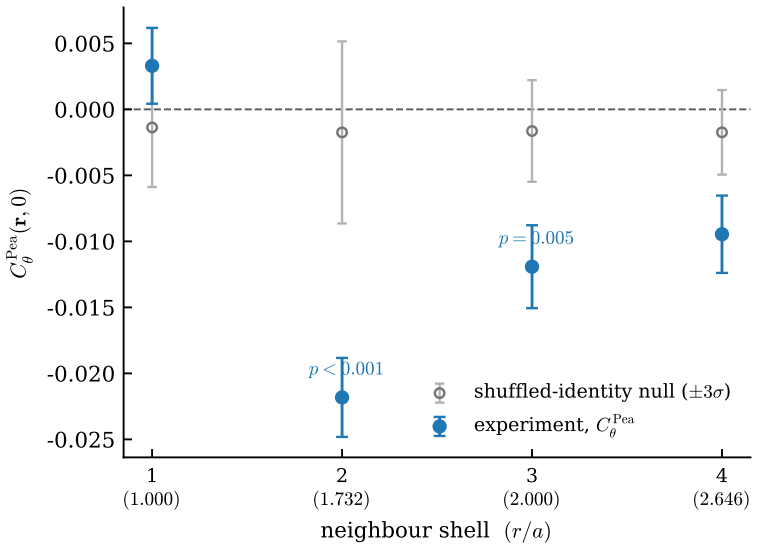


Figure 12: Equal-time Pearson correlation $C_\theta^{\mathrm{Pea}}(\mathbf{r}, 0)$ of the angular field on the first four neighbour shells of the triangular lattice, as a function of the shell index (the corresponding r/a is given below the axis). Filled symbols: experiment, with frame-to-frame standard errors. Open symbols: null model obtained by reshuffling the orientational values among the retained particles of each frame, shown as mean $\pm 3\sigma$. The second and third shells deviate from the null at the $p < 0.001$ and $p = 0.005$ level, respectively, while the first and fourth shells are consistent with it. The magnitudes are small because the coupling is weak relative to the strong rotational noise.

A List of Symbols

Symbol	Meaning
$\theta_j(t)$	Angular coordinate of disk j
$\omega_j(t)$	Angular velocity of disk j, $\omega_j = \dot{\theta}_j$
a	Disk radius
ℓ	Nearest-neighbour lattice spacing, $\ell = 2a + h_0$
h_0	Mean gap width between neighbouring disks
η	Dynamic viscosity of the fluid
I	Moment of inertia of one disk per unit thickness
Γ_{ij}	Dimensionless lubrication coupling coefficient between disks i and j
Γ	Uniform coupling coefficient for all nearest neighbours in a hexagonal lattice
μ	Dimensionless coupling ratio, $\mu = \gamma/\gamma_0$
γ_{bulk}	Bulk rotational damping from the surrounding fluid
T_{ij}^{lub}	Lubrication torque on disk j from the gap with disk i
$T_j^{\mathrm{st}}(t)$	Stochastic torque from thermal fluid fluctuations
$\Lambda_j(t)$	Independent Gaussian white noise on disk j
c	Noise strength, $c\Lambda_j = T_j^{\mathrm{st}}/I$
γ_0	Effective local damping, $\gamma_0 = \gamma_{\mathrm{bulk}}/I + 6\Gamma$
γ	Effective neighbour coupling, $\gamma = \Gamma > 0$ (lubrication coupling; the two facing surfaces of a co-rotating pair shear the gap)
$S(\mathbf{k})$	Triangular-lattice structure factor
$\Gamma(\mathbf{k})$	Mode-dependent damping rate, $\Gamma(\mathbf{k}) = \gamma_0 + \gamma S(\mathbf{k})$
Γ_0	Mean-field effective damping, $\Gamma_0 = \gamma_0 + 6\gamma = \gamma_0 + 6\Gamma$; also bare damping, $\Gamma_0 = \gamma_0 + \gamma S_{\mathrm{gen}}(\mathbf{k})$
$C_\theta(\mathbf{r}, t)$	Equal-time correlation of the angle field, $\langle \theta_{\mathbf{r}}(t)\theta_{\mathbf{0}}(t)\rangle$
$g(R)$	Radial distribution function of the disordered medium
$J(R)$	Microscopic coupling kernel for a single neighbour
$K(\mathbf{R})$	Averaged coupling kernel, $K(\mathbf{R}) = n_0 g(R) J(R)$
n_0	Average two-dimensional number density, $n_0 = N/\Omega$
$S_{\mathrm{gen}}(\mathbf{k})$	Generalized structural function, Fourier transform of $K(\mathbf{R})$
$\Sigma_{\mathrm{dis}}(\mathbf{k})$	Disorder self-energy
$\Gamma_{\mathrm{gen}}(\mathbf{k})$	Generalized damping rate, $\Gamma_{\mathrm{gen}} = \gamma_0 + \gamma S_{\mathrm{gen}} + \Sigma_{\mathrm{dis}}$
$G_0(\mathbf{k}, z)$	Bare propagator, $G_0 = (-iz + \Gamma_0)^{-1}$
$G(\mathbf{k}, z)$	Disorder-averaged propagator
$S_{\mathrm{dis}}(\mathbf{k})$	Structure factor of the displaced lattice, Eq. (5.7)
S_2	Curvature of S_{gen} at $\mathbf{k} = \mathbf{0}$, Eq. (4.20)
Σ_2	Curvature of Σ_{dis} at $\mathbf{k} = \mathbf{0}$, Eq. (4.20)
κ	Curvature of Γ_{gen} at the zone centre, Eq. (4.21)
c_0	Curvature of Γ_{gen} at its minimum, Eq. (4.23)
c_K	Curvature of Γ at the K point, Eq. (3.17)
I_0	Zone integral of $1/\Gamma_{\mathrm{gen}}$, Eq. (4.23)
A	Amplitude of the asymptotic correlation, Eq. (4.23)
$C_{\mathrm{dis}}^{\mathrm{Pea}}(\mathbf{r})$	Disorder-averaged stationary Pearson correlation, Eq. (4.17)
ξ_θ	Correlation length, Eqs. (3.18), (4.22) and (4.24)